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Saving for dowry: Evidence from rural India☆

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ABSTRACT

The ancient custom of dowry, i.e., bride-to-groom marriage payments, remains ubiquitous in many contemporary societies. Using data from 1986–2007, this paper examines whether dowry impacts intertemporal resource allocation and other household decisions in rural India. Utilizing variation in firstborn gender and dowry amounts across marriage markets, we find that the prospect of higher dowry payments at the time of a daughter's marriage leads parents to save more in advance. The higher savings are primarily financed through increased paternal labor supply. This implies that people are farsighted; they work and save more today with payoff in the distant future.

1. Introduction

Marriage payments are an ancient custom that remains widely prevalent in contemporary developing societies (Anderson (2007)). Our objective is to understand how dowry, i.e., transfers from the bridal family to the groom's family at the time of marriage, impacts household decision-making and intertemporal resource allocation.[1] Our geographical setting is rural India where, despite being illegal since 1961,[2] dowry is a nearly universal phenomenon and dowry per marriage often amounts to several years of household income.[3]

It is frequently asserted that Indian parents, who are still primarily responsible for arranging and organizing their children's marriages, start saving for dowry as soon as a daughter is born (Browning and Subramaniam (1995)). Yet, there is little prior empirical evidence on the magnitude of the impact of the institution of dowry on parents' saving behavior. Moreover, the ability to save in advance depends on parents' income constraints, access to formal savings products, and behavioral constraints. A large literature has shown that poverty, limited access to bank accounts, information and knowledge gaps, present-bias, lack of self-control, and limited memory and attention may prevent parents

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[1] Dowry is typically paid in a lump-sum manner at the time of the wedding. The groom's family often receives further transfers from the bride's family after marriage (Bloch and Rao (2002)) but these tend to be substantially smaller in magnitude. The "lumpy" nature of dowry could be due to the couple's inability to divide marital output during the course of a marriage (Becker (1981)) or be driven by the custom of *virilocality* (Botticini and Siow (2003)).

[2] The Dowry Prohibition Act of 1961 prohibits the giving or taking (or the abetting of giving or taking) of dowry in India. The penalty is at least 5 years of imprisonment and a fine no less than INR 15,000 (USD 216) or the value of dowry, whichever is more. The Dowry Prohibition Rules of 1985 made the provisions stricter. More details at http://wcd.nic.in/act/dowry-prohibition-act-1961.

[3] According to REDS (2006), dowry was paid in 95 percent of marriages during 1960–2008.

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from saving in advance for future dowry payments; see DellaVigna (2009), Karlan et al. (2014), and Kremer et al. (2019) for a review. In fact, poverty may exacerbate behavioral biases, reducing poor parents' ability to save even more. Therefore, in this paper we examine whether future dowry payments influence parents' current saving behavior.

Our empirical strategy relies on the fact that, by definition, the dowry expense is higher for parents of a girl relative to parents of a boy.[4] However, we cannot simply compare outcomes after the birth of a girl relative to a boy to estimate the causal impact of future dowry since boy-families and girl-families are likely to differ along other dimensions that are correlated with our outcomes of interest, especially in the Indian context. Therefore, we compare households that differ by firstborn sex. Despite access to prenatal sex-determination technology, the sex ratio at first parity has remained unbiased in India and is frequently used as an exogenous shock in related literature (Das Gupta and Bhat (1997), Visaria (2005), Bhalotra and Cochrane (2010), Anukriti et al. (2016)). We utilize the well-known fact (which we also verify) that Indian parents of a firstborn girl have more girls on average than parents of a firstborn boy (a) due to the presence of son-biased stopping rules[5] (Clark (2000), Bhalotra and van Soest (2008), Jensen (2012), Rosenblum (2013)) and (b) because the former are more likely to practice sex-selective abortions at higher parities (Almond and Edlund (2008), Abrevaya (2009), Bhalotra and Cochrane (2010)).[6] So, instead of comparing boy- and girl-families, we compare firstborn-girl (FG) and firstborn-boy (FB) families, as the former face a higher future dowry expense, on average, due to the greater number of daughters. While it is not obvious that total dowry expense should be higher for parents who have more daughters,[7] in our data, as we show later, an increase in the number of daughters significantly increases total dowry paid by parents and does not affect dowry paid per daughter.

To test if the differences between FG and FB families that we study are due to dowry and not due to other factors, we exploit the variation in expected dowry amounts using a proxy measure of expected dowry. We regress the outcome variables on the interaction of FG dummy and a proxy for expected dowry, thus testing if outcomes vary by expected dowry even within FG families. We assume that parents form expectations about dowry amounts by observing dowries paid by brides or received by grooms in their child's marriage market around the time of their child's birth, where we define a child's marriage market by its caste and state.[8] For a child born in a given year, we construct our proxy for expected dowry as the average net dowry paid by brides or received by grooms from the same caste and state as the child and who married during the year of the child's birth or the prior four years. We also utilize alternate proxies of expected dowry for robustness checks.

We construct a retrospective panel dataset on dowries from the 2006 round of the Rural Economic and Demographic Survey (REDS). Note that, although the first set of interviews in the 2006 REDS were conducted in 2006, 84 percent of the interviews took place in 2008. In addition to data on marriage payments, REDS collected rich information on various forms of savings at the household level and individual labor supply that we utilize. REDS has numerous advantages over other Indian datasets that have been used in the dowry literature that we discuss further in Section 2.

Our findings emphasize the crucial role of traditional cultural institutions like dowry in determining economic behavior. We find that the prospect of higher future dowry increases current savings. In particular, as our proxy for expected dowry increases, FG families significantly increase per capita and total annual savings, both overall and relative to FB families. We also investigate the plausible channels that a household can use to finance the higher savings: (1) lower consumption, (2) increased income, and (3) increased debt. We do not observe any change in total consumption expenditure on items that REDS collected data on. However, we find that fathers of firstborn girls work more days in a year, relative to FB fathers, as anticipated dowry burden goes up, suggesting that at least part of the increased savings is financed through higher income. There is no significant impact on mothers' labor supply, which is not surprising given the low levels of female labor force participation in India (Afridi et al. (2018), Fletcher et al. (2017)). We find some effect on the likelihood of borrowing, but it is not statistically significant across specifications. An average household in our sample appears to be able to save for the bulk of the wedding expenditure in advance.

The increased savings take the form of formal savings in financial institutions, implying that access to formal savings products may not be the binding constraint on savings for an average rural Indian household during the time period of our study—we examine savings in 2007. However, savings in jewelry or precious metals that are traditionally considered an integral part of dowry in India do not increase several years in advance. These patterns are consistent with greater access to financial institutions and instruments in rural India and the less liquid nature of jewelry relative to savings in bank accounts in recent years. Note that this does not imply that jewelry is no longer a significant part of dowry in India. Our finding simply suggests that parents do not save for dowry through advance savings in jewelry in our study-sample and time-period; they may and most likely do utilize the savings in financial institutions to purchase jewelry around the time of marriage. Moreover, jewelry is often transferred across generations, which reduces the need to purchase new jewelry to some extent.

Thus, we find that, on average, parents in rural India work and save more today with payoff after 18 years (the legal minimum age at marriage for girls in India). Unlike decision-making in similar domains where costs are incurred in the present but returns are far out into the future, such as preventive health care, it may be easier for parents to overcome the behavioral constraints when saving for dowry. This is potentially due to better information on the costs and returns, easier mental accounting, lower intra-household disagreement, more frequent nudges and reminders, and greater peer influence in the context of saving for dowry relative to other saving motives. However, the effects on savings and father's labor supply are larger and only significant for households that are above the poverty line. This suggests that more severe income-constraints or behavioral biases resulting from poverty do prevent extremely poor households from saving for their daughters' dowries in advance. This heterogeneity in effects is consistent with the literature on barriers to savings among extremely poor households.

We make a substantial contribution to the literature on marriage payments, especially that on dowry. The bulk of previous work on dowries has been theoretical (e.g., Botticini and Siow (2003), Anderson (2007), Anderson and Bidner (2015)). Empirical research on dowry has been largely limited to marriage market explanations for dowry trends (e.g., Rao (1993), Edlund (2006), Anderson (2007)).[9] In comparison, the literature on the *effects* of dowry has been miniscule (Alfano (2017), Bhalotra et al. (2020), Corno et al. (2020)). Our results are consistent with Deolalikar and Rose (1998) and Rose (2000) who examined the association between female birth, savings, consumption,

[4] Parents of a son may also save for other marriage-related expenses, but, as we show later, the bride's family spends significantly more on the marriage.

[5] A son-biased stopping rule means that parents are more likely to stop childbearing after a male relative to a female birth.

[6] We note that prenatal sex-detection has been illegal in India in public facilities since 1976 and in private facilities since 1994.

[7] For instance, parents who have more daughters may give less dowry per daughter while incurring the same total expense as parents with fewer daughters.

[8] See Section 3 for more details on why this is a reasonable way to define a marriage market.

[9] Logan and Arunachalam (2014) provide a detailed and comprehensive discussion of the dowry inflation debate. Ambrus et al. (2010) examine the emergence of dowry in Bangladesh.

parents' time allocation, and income in India, but they did not show that dowry was the underlying mechanism for their findings. We are the first study that estimates the causal impact of dowry on household savings and parental labor supply in a causal manner.

Our findings are also relevant to the literature on barriers to savings in low-income countries. Household savings are a crucial determinant of welfare, especially in developing countries, where credit and insurance markets are imperfect. Although the relationship between marriage and savings has been previously recognized,[10] we highlight a hitherto under-appreciated motive for savings in dowry-paying societies. Finally, we make a modest contribution to the large literature on income and consumption smoothing (Morduch (1995)). Our finding that households use savings and adjust labor supply to smooth the negative income shock due to dowry is consistent with classical life-cycle and permanent income models (Franco and Brumberg (1954), Friedman (1957), Campbell (1987), Jappelli and Pistaferri (2010)).

2. Data and descriptive statistics

We use the most recent 2006 round of REDS for most of our analysis. REDS was designed to constitute a nationally representative rural panel of Indian households and collects information on labor supply and other economic and demographic variables for 17 major states that contain about 96 percent of India's population. Household- and village-level data were first collected in 1969. Subsequent rounds followed the original households and any split households in 1970, 1971, 1982, 1999, and 2006.

2.1. Sample-selection criteria

We restrict our sample to households that have unmarried individuals under the age of 18 (the minimum age at marriage for women in India) at the time of the survey since this is the group for which our research question is relevant. In addition, we impose several other restrictions that are either necessitated by our research question or the conceptual framework underlying the identification strategy. For instance, we restrict our sample to nuclear families, i.e., families where all children belong to the same set of parents, because financial decision-making in joint families may be very different. As a result, our findings do not apply to joint households. We describe our sample-selection criteria, and why we need to impose them, in Appendix B, where we also show that they do not significantly alter the composition of households in our final estimation sample in terms of caste, religion, below the poverty line (BPL) status, land ownership, household size, and household consumption expenditure relative to the overall 2006 REDS sample.

Although REDS was initially designed to be nationally representative, it is possible that, over time, the sample ceased to be representative of India's rural population. To check if this is the case, we compare our data with the rural sample of the nationally representative 2005 India Human Development Survey (IHDS). We find that our estimation sample is less likely to be Scheduled Caste (SC) and Muslim and more likely to be Other Backward Class (OBC), upper caste, Hindu, and Sikh relative to the 2005 IHDS rural sample. This implies that the average household in our estimation sample is likely to be of higher socioeconomic status (SES) than the average rural Indian household in 2005 due to the correlations between caste, religion, and household SES. To ensure that these differences do not bias our results, we include a rich set of controls in all our regressions (more on this in Section 3).

2.2. Construction of Expected Dowry proxy

Unlike other datasets that record total marriage expenditure in the year of survey by families *similar* to the respondent's family, such as the IHDS, the 2006 REDS collected data on *actual* payments by brides and grooms in the surveyed households.[11] Specifically, it recorded the value of gifts received or given at the time of marriage in addition to the year of marriage and demographic information of spouses (e.g., caste and years of schooling) for 17,401 marriages that occurred during 1986–2007.[12] We compute net dowry as the difference between "gross payments by the bride's family to the groom or his family" and "gross payments by the groom's family to the bride's family," and deflate the nominal amounts using the 2005 national Consumer Price Index.

We plot the distribution of net and gross marriage payments in Fig. 1.[13] The proportion of marriages with a negative net dowry, i.e., where the groom's family paid more to the bride's family than the other way around, is non-zero, but quite small. The vast majority of the marriages involved positive net dowry payments to the groom's family. We do not observe any marriages where the value of gifts was reported to be zero.[14] Fig. A.1 describes the variation in the dowry variable by year of marriage; the top panel shows the raw data on the real net dowry paid by the bride's family and the bottom panel displays, separately, the real value of gifts from the bride's family and from the groom's family.[15]

Ideally, to examine how dowry expectations impact household behavior, we would like self-reported data on how much dowry a family expects to give or receive when their child gets married. Unfortunately, we are unaware of any large-scale dataset that has this information. Consequently, we construct a *proxy* for the true expectations – our *Dowry* variable – by assuming that, after a child is born, parents form expectations about the dowry they will pay or receive for their child in the future by observing marriages that have recently taken place in their reference group, i.e., their marriage market. We assume that the relevant marriage markets are based on caste (and religion) and state,

[10] Horioka (1987) describes that future marriage-related expenses are cited as a significant reason for household savings in Japan. Wei and Zhang (2011), Du and Wei (2013), and Horioka and Terada-Hagiwara (2016) find evidence for marriage-related competitive saving motive in China, the Republic of Korea, and India, wherein parents' savings respond to the marriage market sex ratios faced by their children. Tertilt (2005) examines the relationship between polygyny and savings in Sub-Saharan Africa and Grossbard (2015) theoretically examines how marriage affects employment, savings, and consumption.

[11] In addition to the IHDS, other researchers have used dowry data from the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) and the Status of Women and Fertility (SWAF) surveys. While the ICRISAT data contain retrospective information on marriages, it is only a small survey of 240 households from six villages in three districts of rural South Central India collected in 1983. Although the SWAF survey is relatively new and was conducted in 1993–94, a key shortcoming of it is that it does not report specific dowry amounts and instead provides five ordinal categories that nominal dowries fall into.

[12] The surveys were administered to household heads who provided information on marriages of other household members. Male heads were asked about their non co-resident children, siblings, and non co-resident parents, while female heads were asked about their non co-resident children, and the siblings and non co-resident parents of their husband. One omission that 2006 REDS made is not to collect data on dowries received by co-resident sons of the head. However, since a co-resident son would be married to another head's non-co-resident daughter (typically from the same state and caste given the structure of the Indian marriage market), the dowry information for co-resident sons' marriages may still be in our data.

[13] See Chiplunkar and Weaver (2017) for documentation of the prevalence and evolution of dowry in India using the 1999 round of REDS.

[14] However, our data do contain marriages with missing information on gifts. A detailed discussion of this issue and how we deal with it is in Appendix C.

[15] We also report the composition of dowries, i.e., the kind of things that are usually given as gifts at the time of a daughter's marriage, in rural India using the 2005 IHDS in Table A.1. This information is not recorded in the REDS data.

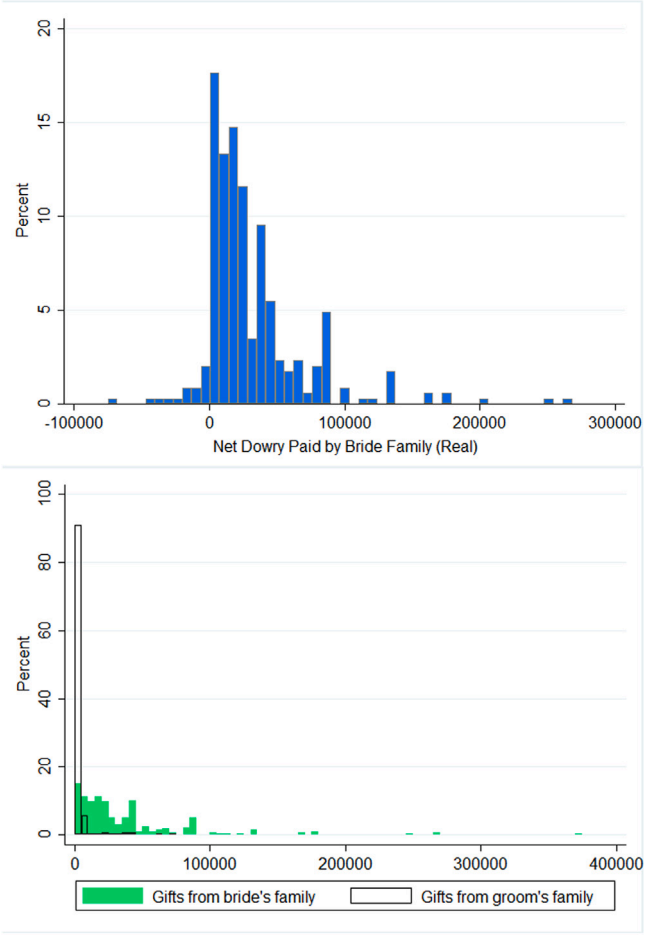


Fig. 1. Distribution of marriage payments (in INR). NOTES: This figure plots the distribution of net dowry (in the top figure) and the distribution of gross payments by the bride's and the groom's families (in the bottom figure) for all marriages in our data that took place during 1986–2007.

and define our proxy for expected dowry as the average net dowry paid by brides or received by grooms from the same caste and state as the child and who married during the year of the child's birth or the prior four years.[16] We trim off the top 1 percent of dowry payments before constructing the expected dowry proxy to prevent outliers from driving our results.

Why we define marriage markets by caste and state: The Indian marriage market is highly endogamous. In the 2005 IHDS, only 4.4 percent of women were married to a spouse from a different caste.[17] In fact, genetic evidence suggests that the practice of endogamy in India is several thousand years old (Moorjani et al., 2013). Although, *jati* is the relevant caste variable for the Indian marriage market, we use 4 broader caste categories: upper castes, SCs, Scheduled tribes (STs), and OBCs. This is because our estimation sample is small relative to the number of *jati*s, resulting in very few observations per *jati*-state–year cell (see Table A1 in the Online Appendix for more details on the number of observations per jati in our estimation sample). Another issue is that the *jati*s reported in datasets such as REDS may not be mutually exclusive, and there may be more measurement error in classification if we group people based on the reported jati names than if we use caste categories such as SC and OBC. Although our caste categories are broad, our objective of using them is simply to distinguish between marriage markets. While it is true that the SCs, for example, do not operate as one marriage market, it is also true that the likelihood of an SC individual marrying another SC individual is substantially and significantly higher than an SC individual marrying an upper-caste individual, making our categorization still useful for a marriage market analysis. To illustrate, in REDS 2006, 91 percent of SC individuals were married to another SC individual. The corresponding numbers for STs, OBCs, and other castes were 93 percent, 95 percent, and 92 percent respectively, implying that our caste categorization captures the marriage markets reasonably well. Also note that although caste is primarily a Hindu phenomenon, the notion of caste-based hierarchy remains well-preserved among many other religious groups in India.[18] Inter-religious marriages are even less common than inter-caste marriages.[19]

Geographically, we define marriage markets by state, rather than by district. This is partly due to data limitations – the average number of marriages in a district–caste–year cell is just four – but mainly because we believe that defining marriage markets by districts is more appropriate for our analysis. Utilizing migration data from the NSS, Fulford (2015) shows that within-state across-district marriage-related migration is non-trivial. Within-state, across-district rural female migration (which is primarily migration at the time of marriage and constitutes the bulk of migration in India) is 24.4 percent for all-India, and there is large variation across states; e.g., 15 percent in Andhra Pradesh and 56.5 percent in Haryana. Defining marriage markets by districts would bias our results as it would affect some states more than others. Moreover, Fulford (2015) shows that across-state migration by females is quite small (less than 1 percent in most cases). Among individuals aged 25 years or older surveyed in the 2008 NSS, 89 percent of women and 93 percent of men lived in the same state where they were born. This is also reflected in the fact that less than 4 percent of the population had moved across states in the last ten years according to the 2001 Census data. Nonetheless, if we lump all *jati*s that have fewer than 100 households into one *jati* and construct expected dowry at the *jati* x State x Year level we find that is significantly correlated with our Caste-group x State x Year dowry measure.

The variation in our key explanatory variable, expected dowry proxy, is at a more aggregate level (caste–state–year) than our savings data, which is at the household level. With 17 states and 22 years, we have 1,285 caste–state–year groupings or marriage markets in our sample.[20] Figs. A.2 and A.3 show the expected dowry variation by year of birth of the firstborn child and by caste groups.[21]

How good is our proxy for expected dowry? To answer this question, we need data on expected dowry. Although REDS or any other large-scale survey has not gathered information on expected dowry, Chari and Maertens (2020) collected such data from three villages in South and West India in 2008. The authors asked parents about dowry forecasts for each of their unmarried children (who on average were 11 years old) and about actual dowry for each of their married children (who on average were 21 years old). They find that the kernel density

[16] Later, for robustness checks, we define our proxy for expected dowry in two alternate ways: by changing the years over which the average is calculated and by using both caste and religion to define the marriage market.

[17] Using responses to matrimonial advertisements in a Bengali newspaper, Banerjee et al. (2013) find evidence in favor of a strong preference for in-caste marriage – e.g., the bride's family is willing to trade-off the difference between no education and a master's degree in the prospective husband to avoid marrying outside their caste. Also see Borker et al. (2017).

[18] For instance, in the 2009 National Sample Survey (NSS), 31 percent of Sikh households identified as being SC.

[19] Nevertheless, expected dowry constructed at the jati x State x Year level is strongly correlated with our current dowry measure: the correlation coefficient is 0.6850 with a p-value = 0.000 (see Fig. A.1 in the Online Appendix).

[20] Due to missing data, the bulk of which are for ST, we do not have all 1,496 cells.

[21] On average, 100 observations are used to construct expected dowry for each caste–state–year of birth cell.

of gross dowries that were actually paid by households of married girls matches remarkably well with the kernel density of the dowry forecasts for unmarried girls. Moreover, they do not find any statistically significant differences between actual and forecast dowry for any of the caste groups and across villages. These results lend some support to our assumption that parents form expectations about future dowry using information on actual dowries and that our proxy for expected dowry is likely to be a good one.[22]

Does our expected dowry proxy predict actual dowry? Our expected dowry proxy is a significant predictor of the actual dowry paid at the time of marriage. Using data on individuals whose year of birth and year of marriage fall within our data, we run an ordinary least-squares (OLS) regression of the actual dowry paid at the time of marriage on the expected dowry proxy in the year of birth. Table A.2 shows that the "first stage" relationship between our expected dowry measure and actual dowry, including state fixed effects, year-of-birth fixed effects, caste fixed effects, and their interactions, is highly significant.[23] Fig. A.4 shows the scatter plot overlaid with the linear prediction plot without any controls. Moreover, the correlation between expected and actual dowry is positive and significant for all four of our caste categories. However, note that what matters for the purposes of our paper is not whether parents' expectations are accurate relative to actual dowry but rather whether our proxy is a good proxy for true expected dowry.[24]

Our expected dowry proxy does not take into account parents' expectations about inflation. However, Chari and Maertens (2020) suggest that households are likely not factoring inflation into their dowry forecasts as their average willingness-to-pay does not increase with child age.[25]

Recall bias in dowry data: Like most survey data, one may be concerned about the recall bias in the reported dowry variable, especially when the year of marriage is too far back in time. To examine the extent of the recall error, we utilize data from the 1999 round of REDS and compare average dowry paid by year of marriage for years that are available in both rounds. Note that the question about dowry was asked differently in both rounds. Nonetheless, Fig. A.5 shows that, for the sample period of our savings analysis (1986–2007), the two rounds report similar dowry amounts. Thus, recall bias is unlikely to be a significant issue for our regression estimates. Given the salience of marriage and dowry in the Indian context, it is unsurprising that the recall error is not substantial.

External validity of dowry data: We corroborate the cross-sectional patterns by caste and religion in 2006 REDS using dowry data from 2005 IHDS. Unlike REDS, IHDS elicits dowry data indirectly by asking respondents how much money is usually spent at the time of the marriage by a groom's or a bride's family that is similar to the respondent's family; this information is only collected for the survey year. Table A.3 shows the sample means of the net dowry paid by the bride's family calculated using these responses for various caste and religious groups. Although our paper focuses on rural India because REDS does not cover urban areas, we also report urban dowries from the IHDS for comparison. Reassuringly, the patterns in Table A.3 are identical to those in the REDS data.[26] The similarities between REDS and IHDS also assuage concerns about differential under- or over-reporting of dowry payments and receipts by the sampled households.

2.3. Savings

Using the detailed information available in REDS, we construct the following measures of household-level saving: savings in financial institutions,[27] savings in jewelry, savings in livestock, market investments,[28] and savings in durable goods. The *flow* saving variables are constructed based on the value of each item purchased (deposits) and sold (withdrawals) during the year before the survey.

The literature describing saving behavior of households in developing countries is quite limited, primarily due to the lack of micro-level data. The bulk of the existing research examines micro-savings for small samples that are not nationally representative (typically, in the context of randomized control trials) or is based on incomplete data on the various forms of savings (see Karlan et al. (2014) for a literature review). For India, we are not aware of any other nationally representative survey that collects detailed data on various forms of savings, except REDS.[29] Information on various types of savings allows us to examine the changes in the entire savings portfolio of a household rather than just one form of saving; we can thus capture substitution across different modes of savings.

Table 1 shows that per capita savings in financial institutions is the biggest component of the annual flow of per capita household savings.

[22] It is useful to think about the determinants of parents' dowry forecasts and whether these factors are being taken into account by our measure. Chari and Maertens (2020) find that dowry forecasts are correlated with household's income, wealth, caste, number of unmarried children, and child ability. Our measure explicitly takes into account caste and, as we later show, our results are robust to controlling for household wealth and the number of children. However, our dowry variable does not take into account child quality–Chari and Maertens (2020) find that expected dowry is increasing in quality for both boys and girls. Even if child quality were a fixed attribute, parents may learn about it gradually and may update their dowry forecasts as their children grow older. Our proxy for expected dowry does not incorporate this learning and assigns each child a proxy for expected dowry determined by his or her year of birth, caste, and state. Thus, (a) our proxy measure of expected dowry is likely to under (over) estimate expected dowry for higher (lower) quality children and (b) the measurement error is likely to be larger for older children at the time of the survey. If higher quality children are more likely to be found in wealthier families that save more, our estimates would bias us towards *not* finding a relationship between expected dowry and savings. Nonetheless, we incorporate year of birth fixed effects quite flexibly in our specification and we interact the dowry variable with firstborn sex which should be uncorrelated with child quality.

[23] We cannot include the firstborn girl interactions here because we do not know the sibling composition and birth order of individuals for whom actual dowry information is available. One may also worry that the significant relationship in Table A.2 is driven by outliers. To test if this is the case, we 'residualized" the fixed effects included in Table A.2 from expected dowry as well as actual dowry and then regressed actual dowry on expected dowry after removing 6 outlier observations—the coefficient estimate remains almost exactly the same as in the last column of Table A.2. Moreover, the relationship remains highly significant if we use logarithmic values of expected and actual dowry (see Table A2 in the Online Appendix).

[24] In Fig. A.2 of the Online Appendix we show that the non-parametric relationship between our proxy for expected dowry and actual dowry is also positive.

[25] Our expected dowry proxy essentially assumes that parents have static or adaptive expectations, i.e., they expect the value of dowry next period to be equal to the weighted average of dowry in the past few periods. If dowry follows a random walk, then the static expectation is also the rational expectation. We explicitly test if the dowry data follows a random walk by performing the modified Dickey–Fuller t test for a unit root in which the series has been transformed by a generalized least-squares regression. The null hypothesis of a unit root or random walk is not rejected for all lags in the model.

[26] Among various questions related to spending at the time of marriage, 2005 IHDS also reports 'average cash' given as a gift at the time of the daughter's marriage. The average cash reported is equal to INR 18,209 (USD 289).

[27] Savings in financial institutions comprises savings in commercial banks, private banks, post office, chit funds, self-help groups, and co-operative societies or banks.

[28] These comprise investments in the stock market, mutual funds, and life insurance.

[29] The Debts and Investment Schedule of NSS collects household data on assets and liabilities but does not include a roster of household members, which is essential for our empirical strategy.

Table 1
Average per capita savings.

Type of saving:	Household per capita saving in 2007 (in 2005 INR)		
	N	Mean flow	Mean stock
	(1)	(2)	(3)
Saving in financial institutions	3,072	573.11	4,306.22
Saving in financial institutions + Cash	3,072	477.59	4,838.41
Saving in financial institutions + Cash + Interest earned	3,072	647.50	5,008.33
Jewelry	3,077	125.20	4,032.46
Market investments	2,255	126.24	964.47
Livestock	1,768	−28.75	3,406.94
Durable goods	3,076	262.17	4,215.28

NOTES: This table reports the average value of different types of savings in per capita terms as reported in 2008 for the year before the survey. The flow values in column (2) are constructed as the difference between the value of each item purchased (deposits) and sold (withdrawals) in the reference year. The stock values in column (3) refer to the value of each item at the time of the survey.

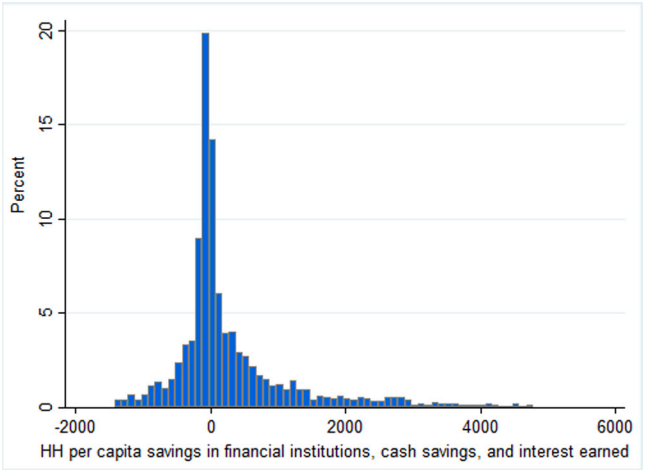


Fig. 2. Distribution of per capita savings in financial institutions (in INR). NOTES: This figure plots the distribution of the flow of per capita household savings in financial institutions, cash savings, and interest earned during 2007 as reported in 2008 (in 2005 INR).

This is true even for the annual stock of savings; however, unlike the flow measure, the magnitude of the stock of jewelry, livestock, and other durable goods is comparable to the stock of savings in financial institutions. This pattern of larger stock but smaller flow of durable goods savings is consistent with the fact that purchases of durable goods tend to be less frequent relative to cash savings. In this paper, we primarily focus on the annual flow of savings; however, later we also briefly discuss the effects on the stock of savings. Average per capita household (flow) savings in financial institutions, cash savings, and interest earned was INR 647 (or USD 10). The average per capita flow of savings across all categories in our data was INR 1,132 (or USD 17) or 19 percent of the per capita household expenditure in 2007 (reported in 2008). Fig. 2 plots the distribution of per capita household savings in financial institutions, cash savings, and interest earned. Fig. A.6 displays how average per capita savings in financial institutions vary across states and caste groups—savings decrease as one moves down the caste hierarchy and there is substantial cross-state heterogeneity in saving amounts. Among upper castes, the states of Gujarat, Kerala, and Punjab have the highest saving rates; these states also rank high in terms of average dowries (see Fig. A.7).

We provide summary statistics for the socioeconomic characteristics of our sample in the first two columns of Table 2. An average household expects to pay or receive INR 26,120 (or USD 412) as dowry.[30] Educational attainment is low—the years of schooling for an average father and mother are, respectively, seven and four. OBCs are the largest caste group in the sample (45 percent), followed by other upper castes (29 percent), SCs (17 percent), and STs (9 percent). In terms of religion, Hindus are the majority (89 percent), followed by Muslims (6 percent), and Sikhs (4 percent). Finally, approximately 43 percent of our sample is BPL. Appendix Table A.4 presents summary statistics for our outcome variables.

3. Empirical strategy

Our goal is to estimate the causal effect of expected future dowry payments on parents' current savings. Although dowry affects parents of boys and girls in the opposite manner, we cannot simply compare boy-families and girl-families as they are likely to differ along other dimensions that are correlated with savings. If girls are born in relatively larger families as compared to boys due to son-biased stopping rules (Clark (2000), Bhalotra and van Soest (2008), Jensen (2012), Rosenblum (2013)), then girl-families would mechanically have lower savings per capita, for instance, irrespective of dowry expectations. Similarly, if sex-selective abortions are more prevalent among groups with certain socioeconomic characteristics that are also correlated with savings, our estimates are likely to suffer from the omitted variables bias. For instance, if richer households are more likely to have sex-selective abortions, girls are more likely to be born in poorer households that likely have lower savings, thereby introducing bias in our estimates. To address these concerns, we instead distinguish between households that differ by *firstborn* sex.

Exogeneity of firstborn sex. Fig. 3 shows that there has been no change in the proportion of females among first births in India over time, despite changes in the availability of prenatal sex-selection technology. Moreover, we do not find significant differences between FB and FG families in terms of socioeconomic characteristics such as expected dowry, caste, religion, father's years of schooling, except for small differences in mother's schooling and being SC (Table 2).[31]

FG families have more girls. Moreover, Indian parents are more likely to practice sex-selective abortions after a firstborn girl (Almond and Edlund (2008), Abrevaya (2009), Bhalotra and Cochrane

[30] The INR 26,120 (or USD 412) amount is roughly similar to the dowries reported in Fig. 1 of Logan and Arunachalam (2014) during 1923–78 and the average net real dowry per marriage in 2007 was equivalent to 14 percent of annual household income.

[31] Anukriti et al. (2016) show that there was no significant change in the sex-composition of first births for any of the socioeconomic groups after ultrasound technology became available. Although the vast majority of Indian families want to have a son if they can only have one child, at a family size of two they prefer having one daughter and one son over having two sons (Jayachandran (2017)). As desired and actual fertility in India are well above one, it is reasonable to assume that parents are not averse to having one girl, despite a strong desire for at least one boy. A formal test of orthogonality, where we regress expected dowry on the FG dummy and several other controls, yields an insignificant and extremely small coefficient on the FG indicator (p-value = 0.39).

Table 2
Differences in socioeconomic status by firstborn sex.

Household variables:	All		Firstborn boy		Firstborn girl		Difference
	N	Mean	N	Mean	N	Mean	
	(1)	(2)	(3)	(4)	(5)	(6)	(7) = (4)–(6)
Expected dowry (in INR 10,000)	3,078	2.612	1,738	2.597	1,340	2.632	−0.035
Father's years of schooling	2,851	6.656	1,608	6.627	1,243	6.692	−0.065
Mother's years of schooling	3,071	3.968	1,736	3.846	1,335	4.127	−0.281*
Father's age	2,855	35.614	1,612	35.635	1,243	35.587	0.048
Mother's age	3,078	31.399	1,738	31.371	1,340	31.434	−0.063
SC	3,078	0.166	1,738	0.148	1,340	0.189	−0.041***
ST	3,078	0.090	1,738	0.096	1,340	0.082	0.014
OBC	3,078	0.452	1,738	0.461	1,340	0.440	0.021
Upper caste	3,078	0.292	1,738	0.295	1,340	0.289	0.006
Hindu	3,078	0.885	1,738	0.890	1,340	0.879	0.011
Muslim	3,078	0.063	1,738	0.059	1,340	0.068	−0.009
Sikh	3,078	0.042	1,738	0.041	1,340	0.043	−0.002
BPL	2,857	0.434	1,625	0.444	1,232	0.421	0.023
Total wealth (PC)	3,078	171,951	1,738	174,729	1,340	168,347	6,382

NOTES: This table provides means of variables used in the analysis. The sample is restricted to households where all children have the same mother. Firstborn boy (girl) refers to households whose firstborn child is male (female). Expected dowry refers to the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. SC, ST, and OBC denote scheduled castes, scheduled tribes, and other backward classes, respectively. BPL denotes a beneficiary of the Below Poverty Line card. Total wealth (PC) refers to the per capita household wealth in land, assets, livestock, jewelry, durable goods, market investments, savings in financial institutions, and cash in hand. *** 1%, ** 5%, * 10%.

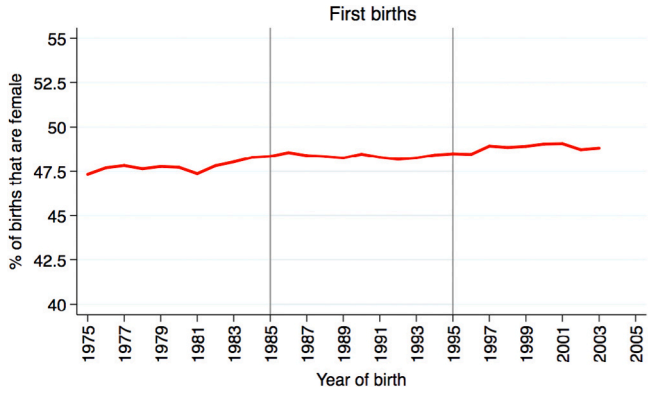


Fig. 3. Evidence against sex-selection at first parity. NOTES: This figure shows the evolution of percent female among first births over time using data from the three rounds of the National Family Health Survey of India. The *y*-axis shows the 5-year moving average of percentage of births that are female. This figure shows that, despite ultrasound availability, the sex ratio of first births has remained normal. The two vertical lines denote the years in which ultrasound availability (a proximate determinant of prenatal sex-selection) underwent structural breaks.
Source: Source: Bhalotra and Cochrane (2010).

(2010), Anukriti et al. (2016)) and follow son-biased stopping rules, resulting in more girls on average in FG relative to FB households. In our data, the average number of girls in FG families is 1.694 as compared to 0.613 girls in FB families. The opposite is true for the average number of boys which is lower in FG families (=0.906) than in FB families (=1.671).

Dowry expense is higher in FG families. This suggests that FG families may face higher total anticipated dowry expense than FB families. However, this may not be the case if parents with more daughters pay sufficiently lower dowry per daughter than families with fewer daughters, or if FG families compensate by having more sons that bring dowry. Our data allow us to test this. Table A.5 shows that the total dowry paid (across all daughters) is higher when a household has more married daughters, and there is no significant effect of the number of daughters on dowry paid per daughter. This is reasonable as a substantially lower dowry offer comes with the risk of not being able to find a suitable groom for the daughter—a highly undesirable outcome for Indian parents. Moreover, as the *net* number of daughters – defined as the difference between the number of daughters and the number of sons – increases, there is a significant increase in the difference between dowries paid for daughters and dowries received for sons. Thus, firstborn gender can be considered a quasi-random dowry shock to the household.

However, simply comparing FG and FB families is also not sufficient. Differential returns from (or costs of) a firstborn-son versus a firstborn-daughter may be due to factors other than dowry, such as higher male productivity in agriculture. FG parents may also save more relative to FB parents to support themselves during old age, as the custom of patrilocality implies that daughters (and not sons) move out of the natal home upon marriage. To explicitly take into account the contribution of dowry expectations to savings, we, therefore, interact the FG dummy with the expected dowry variable. This allows us to control for the main effect of FG, that will capture any changes in per capita saving due to firstborn sex that could result from factors unrelated to dowry; for instance, higher fertility among FG families due to the desire for at least one son.

Ideally, we would like to use data on household savings before the birth of the first child and compare it with savings afterwards. However, 2006 REDS only reports cross-sectional information on savings. Note, however, that REDS is a panel survey, and the same households were also surveyed in 1999. Unfortunately, the survey questions were not exactly the same in both rounds, making it difficult to use the 1999 round for a panel-specification without significant measurement error.

To investigate whether FG parents save more than FB parents due to expected future dowry payments, we estimate the following specification for mother i from caste c in state s and whose first child was born in year t:

$$\begin{aligned} Saving_{icst}^{2008} = {} & \alpha + \beta_1 FirstGirl_i \times Dowry_{cst} + \beta_2 Dowry_{cst} + \beta_3 FirstGirl_i \\ & + \pi_{st} + \phi_{ct} + \psi_{sc} + \eta_c FirstGirl_i + \eta_s FirstGirl_i + \eta_t FirstGirl_i \\ & + \omega_c + \delta_s + \theta_t + \mathbf{X}_i'\gamma + \epsilon_{icst}, \end{aligned} \tag{1}$$

where $Saving_{icst}^{2008}$ denotes various measures of household saving last year reported in 2008; $FirstGirl_i$ indicates that the firstborn child is female; $Dowry_{cst}$ is expected dowry defined as the average dowry paid by brides from caste c in state s.[32] who were married during the year of

[32] State reflects the state of residence at the time of survey, and not necessarily the state of marriage or the state of first birth. However, this is unlikely to be a major source of measurement error due to low inter-state migration in India, especially after marriage.

the child's birth or the prior four years (i.e., during $t, t-1, t-2, t-3, t-4$);[33] $\mathbf{X}_i$ is a vector of covariates comprising parents' years of schooling and ages at the time of survey, and indicators for religion and month of survey.[34] We report unweighted regressions in the main set of tables. However, our results remain the same when we use weights.[35] Standard errors are clustered at the state level. We also compute standard errors that are wild-cluster bootstrapped by state as a robustness check.

The coefficient β_2 captures how savings in FB families respond to expected dowry receipts. The coefficient of primary interest is β_1 which captures the differential response of FG families to expected dowry, relative to the response of FB families. To exclude other confounding factors related to the caste, state, gender, and year of birth of the firstborn child, we control for all main and interaction fixed effects for these factors. For instance, if certain castes are richer and therefore save more, the caste fixed effects (ω_c) would capture this confounding variation. If there is state or time variation in wealth status of various castes that can explain the differential saving behavior, it would be captured by the state–caste and caste–year fixed effects (ϕ_{ct} and ψ_{sc}). The state–year fixed effects (π_{st}) take into account any differential trends across states in economic prosperity. If the fertility response to a firstborn girl differs across castes or states or time due to differential son preference, it would be captured by $\eta_c FirstGirl_i$, $\eta_s FirstGirl_i$, and $\eta_t FirstGirl_i$.

Given the fixed effects included in specification (1), our coefficients of interest are identified from (a) changes in expected dowry payments within state–caste groups over time, (b) differences in expected dowry payments across castes within a state–year, and (c) differences in expected dowry payments across states within a caste–year. The literature on dowry variation in India suggests a few potential sources of such variation. During our study period (1986–2007), the states of Andhra Pradesh (1986), Tamil Nadu (1989), Maharashtra (1994), and Karnataka (1994) amended the laws governing female inheritance rights for Hindus, Buddhists, Jains, and Sikhs, thereby generating variation in bequest-motive for dowry by state–year of birth–religion. Roy (2015) has shown that these amendments affected dowry payments. Another potential source of variation is macroeconomic changes such as those brought about by trade liberalization. Using an older (1999) round of REDS, (Chakraborty, 2015) has shown that tariff cuts resulting from the 1991 trade reforms in India also affected dowries differentially across districts and castes/ religions via effects of tariff cuts on within-caste wealth inequality across districts, consistent with the theory proposed by Anderson (2003). Chiplunkar and Weaver (2017), also using the 1999 REDS data, demonstrate that the dowry variation in India during 1975–1999 can be explained by greater differentiation in groom quality over time and marriage market competition within state–caste-religion for more educated grooms; this explanation analyzes dowry as groom-price. However, given the small size of our estimation sample, we do not explicitly model or utilize the dowry variation generated by these channels.

Any remaining threats to identification of the coefficient of interest, β_1, as the causal effect stem from omitted caste–state–year specific factors that may be correlated with $Dowry_{cst}$ and that differentially affect FG and FB parents. To address the former concern, we replace the main effect of $Dowry_{cst}$ with caste × state × year fixed effects in specification (1) to estimate an even stricter specification that non-parametrically controls for everything that varies at the caste–state–year level and is correlated with household savings.

$$\begin{aligned} Saving_{icst}^{2008} = {} & \alpha + \beta_1 FirstGirl_i \times Dowry_{cst} + \delta_{cst} + \beta_3 FirstGirl_i \\ & + \pi_{st} + \phi_{ct} + \psi_{sc} + \eta_c FirstGirl_i + \eta_s FirstGirl_i + \eta_t FirstGirl_i \\ & + \omega_c + \delta_s + \theta_t + \mathbf{X}_i'\gamma + \epsilon_{icst}, \end{aligned} \tag{2}$$

4. Effect on savings

In Table 3 we present results from Eqs. (1) and (2) that estimate the impact of expected future dowry payments on parents' current saving behavior. We expand the definition of the savings variable as we move from column (1) to column (6). In the first two columns, the outcome is annual flow of per capita saving in financial institutions; in the next two columns we add annual per capita flow of cash saving to the first outcome; and in the last two columns we add the annual per capita flow of interest earned by the household on its savings to the second outcome. Note that the expected dowry variable, unless otherwise mentioned, is in INR 10,000.

The coefficient of *Firstborn girl* is negative and always insignificant, implying that, in the absence of dowry expectations, there is no difference between the yearly per capita saving amount in FG families and FB families. However, as the interaction coefficient demonstrates, when expected dowry is positive, FG families save significantly more than FB families on a per capita annual basis, and, within FG families, per capita annual savings increase with the amount of expected dowry. The specifications without caste–state–year fixed effects show that FB families do not significantly alter per capita savings when anticipated dowry receipts are higher, as the coefficients of *Expected Dowry* are insignificant. As per column (6), if expected dowry increases by INR 10,000, annual per capita savings go up by INR 617.54 in FG families, on average. For the mean level of expected dowry (=INR 26,120), FG families save INR 1613 (=617.54*2.612) more on per capita basis each year than FB families.

Table A.6 shows a more detailed version of Table 3 and highlights the importance of controlling for various omitted factors correlated with firstborn sex, dowry, and savings in order to narrow down the causal effect of interest.[36] Our results are driven by Hindu households (Table A.7), which is not surprising as Hindus comprise the bulk of our sample. The results are similar if we use *total* annual household savings as opposed to per capita savings as the outcome variable (Table 4). The last column of Table 4 shows that if expected dowry increases by INR 10,000, total household savings, on average, go up by INR 3789.41 in FG families. As we use a flow measure of saving, it is also reassuring that the results remain the same when we control for the stock of

[33] The robustness checks using alternate definition of dowry expectation are provided in Section 7.

[34] Replacing parents' years of schooling with indicators for primary, secondary, and higher school completion does not alter our results; these robustness checks are available in Online Appendix Table A3.

[35] The 2006 REDS data do not provide sampling weights; hence we construct them in the following manner, based on Andrew Foster's suggestion. Using the village listing data which includes all households in REDS villages, we create an indicator for the households that are actually sampled and regress it on the observables in the listing data. These inverted predicted probabilities serve as weights, assuming that the observables capture differential reasons for being surveyed. The observables in the listing sheet data used to construct weight are household size, number of earners in the household, head's age, head's years of schooling, indicators of head's caste (SC, ST, OBC, upper caste), religion (Muslim), and gender, and state fixed effects.

[36] For instance, the inclusion or exclusion of Firstgirl x State FE in columns (6)–(7) of Table 3 makes a big difference to the interaction coefficient of interest. This is expected as the impact of firstborn girl on various outcomes, such as fertility, is likely to differ across states due to substantial state-level heterogeneity in India, and omitting these variables can bias our estimates. If firstborn-girl families in more gender-biased states have higher fertility, this would mechanically lower per capita savings. More gender-biased states may also be more likely to have higher dowry for various reasons; for example women may be less likely to work or have lower education in these states, causing higher bride-to-groom transfers on the marriage market. If so, excluding Firstgirl x State FE will bias our coefficient of interest in the downward direction.

Table 3
Impact of expected dowry on the flow of household per capita saving.

Dependent variable:	Household per capita saving in 2007					
	Saving in financial institutions		Plus cash saving		Plus interest earned	
	(1)	(2)	(3)	(4)	(5)	(6)
*Firstborn girl * Expected dowry*	419.24 [190.02]** (239.26)*	556.25 [271.31]* (262.68)***	395.89 [190.15]* (245.13)*	485.14 [251.08]* (242.62)***	507.69 [231.68]** (290.19)*	617.54 [304.20]* (305.86)***
Expected dowry	60.86 [306.28]		−98.08 [316.61]		−113.09 [339.35]	
Firstborn girl	−406.13 [1728.71]	−639.85 [3342.15]	−1044.67 [1584.76]	−1132.86 [3124.21]	−1942.95 [1912.26]	−2168.65 [3488.94]
N	2,840					
Dep var mean for Firstborn boy	745.95		670.20		874.69	
$\mathbf{X}_i$	x	x	x	x	x	x
Caste FE	x	x	x	x	x	x
YOB FE	x	x	x	x	x	x
State FE	x	x	x	x	x	x
State*YOB FE	x	x	x	x	x	x
Caste*YOB FE	x	x	x	x	x	x
Caste*State FE	x	x	x	x	x	x
Firstborn girl*YOB FE	x	x	x	x	x	x
Firstborn girl*State FE	x	x	x	x	x	x
Firstborn girl*Caste FE	x	x	x	x	x	x
Caste*State*YOB FE		x		x		x

NOTES: This table reports the coefficients from specifications (1) and (2) estimated for households where all children have the same mother. Each column is a separate regression. The dependent variable is the flow of per capita household saving in financial institutions in columns (1) and (2), plus per capita cash saving in columns (3) and (4), and plus per capita interest earning in the last two columns. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Caste refers to indicators for SC, ST, OBC, and upper castes. YOB refers to the year of birth of the firstborn child. State refers to the state of residence at the time of survey. $\mathbf{X}_i$ controls for parents' age, schooling, religion, and month of survey. Standard errors in brackets are clustered by state and wild-cluster bootstrapped errors by state are in parentheses. *** 1%, ** 5%, * 10%.

savings at the beginning of 2007 (column (1) of Table 5). Additionally, if we construct our proxy for expected dowry using dowry data from a different caste group in the same state as the child, the interaction term in Table 3 becomes insignificant (see Appendix Table A.8). This finding lends support to our definition of marriage markets.[37]

Together, the results presented in this section suggest that the impending future lump-sum dowry expense induces FG families to start saving more in advance.

4.1. Fertility as a potential confounder

Although specifications (1) and (2) control for a large number of omitted variables, one may still worry about confounding factors that vary by caste–state–year and that differentially affect FG and FB parents—i.e., these are omitted variables correlated with our interaction term. One such factor could be fertility. If $Dowry_{cst}$ is positively correlated with son preference at the caste–state–year level, our interaction term will be negatively correlated with fertility, introducing upward bias in the estimated effect of expected dowry. A similar bias can result if higher $Dowry_{cst}$ causes more sex-selection in FG families as they try to have a compensating son who would receive dowry, lowering future fertility. A higher sex ratio may also reduce $Dowry_{cst}$ due to the scarcity of women on the marriage market, making parents *less* likely to save and not more.

To address these endogeneity concerns related to differential fertility and sex-selection in FG and FB families, we first show that our results survive controlling for indicators for the number of children (column (2) of Table 5) and controlling for the proportion of sons (Table 6).[38] Second, we explicitly estimate the effect of expected dowry on differential fertility and sex-selection behavior in FG relative to FB families. To do so, we use data on fertility and sex-selection from the 1998–99 and 2005–06 rounds of the Demographic Health Survey of India, also known as the National Family Health Survey (NFHS).[39] We restrict our analysis to the rural sub-sample of NFHS since our dowry data from REDS does not cover urban India. We prefer using NFHS for this analysis as it reports complete birth histories, including the dates of live births and of any child deaths, for each surveyed woman, where as

[37] Our results are also robust to controlling for per capita value of the durable goods owned by the household, household size, the interaction between a vector of family characteristics and our dowry variable, and variance in dowry, and to replacing our dowry variable with its logarithm, to removing outliers from the savings variables, and to replacing our dowry variable with its natural logarithm. To check if savings are differentially affected by the volatility in recent dowries among firstborn-boy and firstborn-girl families, we replace our dowry variable with the variance of dowry variable and find that the interaction term is positive but always insignificant. These results are available in Tables A3, A4, A5, A6, A7, and A8 of the Online Appendix.

[38] In Table A.9, we re-estimate the effects on savings for families that only have one child. The sub-sample of one-child families offers a strict test for our story since the saving behavior of these families has not yet been affected by the differential likelihood of higher parity births or sex-selection of these births by firstborn sex (given that we flexibly control for duration since first birth effects). Despite the small sample size, we find that families that have only a girl child save significantly more than families that have only a boy child for given expected dowry. The higher magnitude of the savings effect in Table A.9 relative to Table 3 suggests that per capita savings fall as the number of children increases; however, given the small sample size, these coefficients are less reliable.

[39] NFHS interviewed women aged 15–49 at the time of the survey. We do not use the 1992–93 round of NFHS as it did not distinguish upper castes from other backward classes (OBCs)—which is essential for our identification strategy.

Table 4
Total (instead of per capita) household saving.

Dependent variable:	Total Household Saving in 2007					
	Saving in financial institutions		Plus cash saving		Plus interest earned	
	(1)	(2)	(3)	(4)	(5)	(6)
*Firstborn girl * Expected dowry*	2,527.05**	3,069.96*	2,457.32**	2,895.15**	3,176.43**	3,789.41*
	[933.70]	[1,460.11]	[929.53]	[1,288.65]	[1,318.99]	[1,877.15]
Expected dowry	450.71		−224.83		−436.14	
	[1,678.00]		[1,650.52]		[1,841.82]	
Firstborn girl	−3,798.81	−5,508.50	−7,405.35	−9,032.55	−12,743.13	−15,613.02
	[9,699.30]	[17,428.85]	[8,999.26]	[16,102.43]	[11,315.43]	[19,124.74]
N	2,840					
$\mathbf{X}_i$	x	x	x	x	x	x
Caste FE	x	x	x	x	x	x
YOB FE	x	x	x	x	x	x
State FE	x	x	x	x	x	x
State*YOB FE	x	x	x	x	x	x
Caste*YOB FE	x	x	x	x	x	x
Caste*State FE	x	x	x	x	x	x
Firstborn girl*YOB FE	x	x	x	x	x	x
Firstborn girl*State FE	x	x	x	x	x	x
Firstborn girl*Caste FE	x	x	x	x	x	x
Caste*State*YOB FE		x		x		x

NOTES: This table reports the coefficients corresponding to specifications (1) and (2) estimated for households where all children have the same mother. Each column is a separate regression. The dependent variable is total household saving in financial institutions in columns (1) and (2), plus cash saving in columns (3) and (4), and plus interest earning in the last two columns. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Caste refers to indicators for SC, ST, OBC, and upper castes. YOB refers to the year of birth of the firstborn child. State refers to the state of residence at the time of survey. $\mathbf{X}_i$ controls for parents' age, schooling, religion, and month of survey. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

Table 5
Robustness checks for the effect on savings.

Dependent variable:	Household per capita saving in 2007	
	(financial institutions + cash + interest earned)	
	Additional controls:	
	Saving deposits at the start of 2007 (1)	FE for #children (2)
*Firstborn girl * Expected dowry*	495.38*	612.76*
	[281.47]	[211.71]
Firstborn girl	−512.63	−2140.94
	[3464.00]	[3522.45]
N	2,840	2,840
Dep var mean for Firstborn boy	874.69	874.69

NOTES: This table reports the coefficients for specification (2) controlling for saving deposits at the start of 2008 in column (1) and fixed effects for the number of children in column (2). The sample is restricted to households where all children have the same mother. The dependent variable is the sum total of per capita savings in financial institutions, cash savings, and interest earned in 2007. Each column corresponds to a different regression. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Dep var mean is the mean of the dependent variable for firstborn boy households. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

in REDS we need to impute birth year and birth order from children's ages. Moreover, NFHS is substantially larger in sample size than REDS.

We estimate the following equation by using the number of children and the proportion of sons among second and higher parity births at the time of survey for mother i from caste c in state s and whose first child was born in year t as the dependent variables:

$$Y_{icst} = \alpha + \beta_1 FirstGirl_i \times Dowry_{cst} + \beta_2 Dowry_{cst} + \beta_3 FirstGirl_i + \pi_{st} + \phi_{ct} + \psi_{sc} + \omega_c + \delta_s + \theta_t + \mathbf{X}_i'\gamma + \epsilon_{icst}, \quad (3)$$

The vector $\mathbf{X}_i$ comprises parents' years of schooling, parents' ages, and indicators for religion, month of survey, and household standard of living at the time of survey. The inclusion of the FG main effect allows us to control for any changes in fertility or sex-selection due to firstborn sex that could result from factors unrelated to dowry; for instance, higher fertility among FG families due to the desire for at least one son. If certain castes are richer and therefore can more easily afford sex-selection technology, the caste fixed effects (ω_c) would capture this confounding variation. If there is state or time variation in wealth status of various castes that can explain the differential fertility behavior, it would be captured by the state–caste and caste–year fixed effects (ϕ_{ct} and ψ_{sc}). The state–year fixed effects (π_{st}) take into account any differential trends across states in factors that affect fertility, son preference, and access to sex-selection technology. Like our savings specification, we also replace the main effect of $Dowry_{cst}$ with caste × state × year fixed effects to estimate an even stricter specification that non-parametrically controls for everything that varies at the caste–state–year level and is correlated with fertility or sex-selection.

Table 7 presents estimates from specification (3). The coefficients of *Firstborn girl* imply that FG families have more children and practice

Table 6
Robustness check for the effect on savings: Controlling for the proportion of sons.

Dependent variable:	Household Per Capita Saving in 2008					
	Saving in financial institutions		Plus cash saving		Plus interest earned	
	(1)	(2)	(3)	(4)	(5)	(6)
*Firstborn girl * Expected dowry*	415.241**	556.159*	388.075*	479.043*	497.124**	609.212*
	(188.352)	(273.204)	(186.939)	(252.785)	(225.458)	(301.575)
Expected dowry	64.385		−91.194		−103.773	
	(305.886)		(314.327)		(335.142)	
Firstborn girl	−489.111	−641.473	−1,206.605	−1,246.842	−2,162.101	−2,324.344
	(1,691.462)	(3,291.391)	(1,555.138)	(3,074.922)	(1,923.838)	(3,480.122)
N	2,840					
$\mathbf{X}_i$ + Proportion of sons	x	x	x	x	x	x
Caste FE	x	x	x	x	x	x
YOB FE	x	x	x	x	x	x
State FE	x	x	x	x	x	x
State*YOB FE	x	x	x	x	x	x
Caste*YOB FE	x	x	x	x	x	x
Caste*State FE	x	x	x	x	x	x
Firstborn girl*YOB FE	x	x	x	x	x	x
Firstborn girl*State FE	x	x	x	x	x	x
Firstborn girl*Caste FE	x	x	x	x	x	x
Caste*State*YOB FE		x		x		x

NOTES: The only difference between this table and Table 3 is the addition of a new control: the proportion of sons.

Table 7
Impact of expected dowry on fertility and sex ratio, NFHS rural data.

	(1)	(2)	(3)	(4)	(5)
Dependent variable:	Number of births				
*Firstborn girl * Expected dowry*		0.008	0.002	0.003	0.004
		[0.015]	[0.012]	[0.013]	[0.013]
Expected dowry		−0.084	−0.004	−0.004	
		[0.057]	[0.037]	[0.016]	
Firstborn girl	0.273***	0.249***	0.287***	0.287***	0.284***
	[0.032]	[0.050]	[0.039]	[0.038]	[0.039]
N	60,248	60,248	59,737	59,737	59,737
Dep var mean for Firstborn boy	3.095	3.095	3.095	3.095	3.095
Dependent variable:	Fraction sons (parity ≥ 2)				
*Firstborn girl * Expected dowry*		0.008*	0.008*	0.007	0.007
		[0.004]	[0.004]	[0.004]	[0.004]
Expected dowry		−0.0002	−0.005**	−0.008***	
		[0.002]	[0.002]	[0.003]	
Firstborn girl	0.030***	0.008	0.009	0.009	0.010
	[0.004]	[0.010]	[0.010]	[0.010]	[0.010]
N	51,243	51,243	50,810	50,810	50,810
Dep var mean for Firstborn boy	0.530	0.530	0.530	0.530	0.530
$\mathbf{X}_i$			x	x	x
Caste FE + YOB FE + State FE			x	x	x
State*YOB FE				x	x
Caste*YOB FE				x	x
Caste*State FE				x	x
Caste*State*YOB FE					x

NOTES: This table reports the coefficients corresponding to specification (3). Each column is a separate regression. The dependent variable in the top panel is the total number of births and in the bottom panel is the proportion of male births among second and higher parity births. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Dep var mean is the mean of the dependent variable for firstborn-boy households. YOB refers to the year of birth of the firstborn child. $\mathbf{X}_i$ controls for parents' age, years of schooling, religion, and household standard of living at the time of survey. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

greater sex-selection at second and higher parities even if expected dowry is zero. The interaction coefficients are, however, highly insignificant for the fertility regressions implying that there is no significant differential effect of dowry by firstborn sex on future childbearing. For sex-selection, the interaction terms are also insignificant at conventional levels, but not as strongly as in the case of fertility. We obtain similar results from REDS data; see Table A.10.

These findings not only assuage concerns about endogenous fertility, but are also important in their own right. It is frequently claimed that dowry is an underlying cause of son preference, male-biased fertility, and discrimination against girls in India. Bhalotra et al. (2020), for instance, find that an unexpected one-time increase in the price of gold led to immediate rise in fetal and infant mortality of girls presumably because households perceived the gold price shock as an increase in the value of dowry. Similarly, Alfano (2017) finds that an amendment that made the Indian anti-dowry law stricter in 1985 led to decreases in male-biased fertility behaviors as it potentially made the dowry cost of daughters smaller. If dowry indeed causes male-biased behaviors, by the same reasoning, FG families should also be more likely to practice sex-selection at higher parities as expected dowry rises. However, we do not find strong evidence for this, with the caveat that our specification is quite different from those used in prior work. Our findings are consistent with Borker et al. (2017), who argue that dowry is not the root cause of sex-selection in India; instead, sex-selection is driven by marriage-market frictions that arise from the way the institution of marriage is structured in India.

While the desire for at least one son is real, and affects childbearing decisions in India (Jayachandran (2017)), we find that it leads to higher fertility and higher sex ratios *even in the absence of dowry*, but dowry does not seem to be an additional significant explanatory factor. This finding is reasonable for the following reason. If dowry expenditure was the predominant reason for sex-selective behavior, one would expect parents to always prefer a son over a daughter, irrespective of birth parity and sex-composition of other children, since all daughters require a dowry. However, the pattern of son preference and sex-selection in India is highly parity-specific and depends on the number of sons a family has. Indian parents seem to strongly desire *one* son, and conditional on having at least one son, stated preferences are quite gender neutral.[40] The bulk of sex-selective abortions take place in families that do not have a son.

5. How are savings financed?

Next we test if FG families make adjustments to consumption expenditure, labor supply, and debt to finance the higher savings that we observe above.

Consumption. For per capita consumption expenditure, we estimate specification (2)—the coefficient of *Firstborn girl* in Table A.11 is negative and often significant suggesting that FG families consume significantly less per capita than FB families even when dowry is not a consideration. This could be because FG families have more children than FB families and thus have more household members. But the interaction term is never significant, implying that there is no differential impact of expected dowry on per capita food, non-food, as well as total consumption expenditure.

Labor supply. The birth of a first child can affect parents' time allocation in the following ways. As the permanent dowry-related income shock due to firstborn sex is a pure lottery and does not change the reward or wage from working, there is no substitution effect. The income effect of future dowry implies that, in the absence of credit constraints, FB parents should increase leisure (decrease labor supply) and FG parents should increase labor supply. However, if the household is credit constrained, current labor supply may not decrease for FB parents despite higher permanent income. If FG parents are income-constrained, they may also increase labor supply in an attempt to supplement their income to finance the higher savings needed for the future dowry expense. We focus on father's labor supply in this paper as 89 percent of the mothers in our dataset report being a housewife as their primary occupation.[41]

The 2006 REDS provides information on the number of days worked each year between 1986 and the year of survey for each household member. We construct a panel dataset of fathers' labor supply from this information and estimate the effect on father i from caste c in state s in year t' and whose first child was born in year t using the following specification:

$$\begin{aligned} L_{it'} = {} & \alpha + \beta_1 FirstGirl_i \times Post_{t'>t} \times Dowry_{cst} \\ & + \beta_2 Post_{t'>t} \times Dowry_{cst} + \beta_3 FirstGirl_i \times Post_{t'>t} \\ & + \delta_{st'} + \theta_{ct'} + \pi_{tt'} + \gamma_i + \omega_{t'} + \epsilon_{it'}, \end{aligned} \tag{4}$$

where $L_{it'}$ denotes the number of days worked in year t'; $Post_{t'>t}$ equals 1 if $t' > t$, and 0 otherwise; and $FirstGirl_i$ and $Dowry_{cst}$ are defined as before. We include time interaction fixed effects (i.e., $\delta_{st'}, \theta_{ct'}, \pi_{tt'}$) as well as time fixed effects ($\omega_{t'}$) in this specification. The coefficient β_2 captures how expected dowry affects father's number of days worked after the birth of a firstborn boy and β_1 captures the differential response of parents of a firstborn girl after her birth. The panel nature of the labor supply variable allows us to also control for father fixed effects (γ_i).

Column (1) in Table 8 shows that greater future dowry expenditure causes FG fathers to work more relative to FB fathers after the birth of their first child; the latter do not exhibit a significant change in their annual labor supply. Note that, in the absence of dowry expectations, FG fathers do not work more than FB fathers after their first child is born. For mean expected dowry (=INR 26,120), the triple-interaction coefficient (=3.64) translates into roughly 10 more days worked each year for an FG father relative to his labor supply before the daughter's birth and relative to an FB father. Thus, FG parents attempt to finance the higher savings through higher earnings. Appendix Table A.12 shows that the labor supply adjustment takes place primarily on the intensive margin.

Adjustment along the migration margin is small as reflected in the fact that only 1.39 percent of the household heads in our sample were surveyed in a village different from the village where they were born (or married into for female heads). Similarly, only 2.21 percent of the fathers were migrants at the time of the survey.

Borrowing. In Panel A of Table 9, we examine if there is also a corresponding rise in debt for FG families relative to FB families due to expected dowry. Column (1) shows that for average expected dowry (=INR 26,120), FG families are 6 percentage points more like to have borrowed the year before the survey, although we lose significance once caste–state–year fixed effects are included. In the absence of dowry expectations, FG families are *less* likely to have borrowed than FB families. Each INR 10,000 rise in expected dowry raises the likelihood of borrowing in a year by 3 percentage points for FG families.

Like employment, REDS also has data on the borrowing history of each household. However, more than 90 percent of the loans in the data were taken out during seven years before the survey, suggesting that there is serious recall bias in the historical loan data. Keeping this caveat in mind, we use specification (4) to estimate the effect on total amount borrowed by FG families each year, relative to FB families, to finance future dowry. We restrict the sample to 1980–2008

[40] The emphasis on having one son is likely due to the lack of institutional old-age support (Ebenstein and Leung (2010)) and cultural norms such as patrilocality (Ebenstein (2014)) and male-centered funeral rituals.

[41] The predictions for mother's labor supply are not straightforward. While child rearing may involve some decline in market work irrespective of child gender, returns to investment of women's time in child-care on the child's marriage market may be an additional consideration while allocating time.

Table 8
Impact of expected dowry on father's labor supply.

Dependent Variable:	Father's days worked in a year					
	All	Above poverty line	Below poverty line	Firstborn ≤ 5	Firstborn ≤ 10	Firstborn ≤ 15
	(1)	(2)	(3)	(4)	(5)	(6)
*Firstborn girl * Post* Expected dowry*	3.64**	5.54**	2.19	1.62	4.64	4.39**
	[1.66]	[1.95]	[3.94]	[4.10]	[3.96]	[1.85]
*Expected dowry * Post*	0.89	-0.24	-0.66	-1.21	-3.20*	-0.64
	[2.49]	[3.17]	[4.38]	[3.80]	[1.71]	[2.06]
*Firstborn girl * Post*	-9.85*	-14.49**	-11.32	-0.70	-7.68	-11.97*
	[5.29]	[6.77]	[11.77]	[11.14]	[10.25]	[6.54]
N	71,282	36,703	29,395	19,876	37,852	58,433
Dep var mean for Firstborn boy	156.30	146.76	167.76	95.82	119.23	142.39

NOTES: This table reports the coefficients corresponding to specification (4). The sample is restricted to households where all children have the same mother. Each column corresponds to a different regression. The dependent variable is the number of days worked each year. *Firstborn girl* indicates that the firstborn child of the household is female. Post indicates that the year of labor is later than the first child's year of birth. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Dep var mean is the mean of the dependent variable for firstborn boy households. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

to minimize recall bias. Panel B of Table 9 shows that the triple-interaction coefficient is positive (though insignificant) implying that borrowing post-birth is larger in FG families as expected dowry rises.

On the whole, we find that FG families finance the increased savings after the birth of their daughter by increasing fathers' labor supply and by borrowing somewhat more.

6. Discussion

The birth of a daughter in dowry-paying societies, such as India, implies an almost certain increase in parents' future expenditure. Faced with such a lumpy expense, one way parents of daughters could finance dowries is through borrowing around the time of marriage. Indeed, in our data, households are more likely to borrow during the year of marriage than otherwise. The likelihood that a household applied for a loan (formal or informal) in the year before the survey is significantly higher if a marriage took place in the household during the same year than if it did not (47 percent versus 18 percent). The corresponding numbers for borrowing specifically for marriage or ornaments are 32 percent in households where a marriage took place last year versus 0.54 percent in households where no marriage occurred last year.[42] However, parents may be unable to borrow the entire amount necessary for dowry and other marriage-related expenses at the time of marriage due to credit constraints that are likely to be substantial in rural India.

An alternative way to raise liquidity for dowry is, therefore, saving in advance. Saving for dowry was prevalent even in medieval Florence. In 1425, Cosimo de' Medici established *Monte Delle Doti* or Dowry Bank, a public fund that sought to assist families in accumulating adequate dowries. The fund accrued a guaranteed interest rate on deposits made upon the birth of a daughter that could be withdrawn after an agreed number of years to become the daughter's dowry (Frick (2011), Strathern (2015), Franklin (2017)). Our results above have established that, indeed, households that face a higher dowry burden in rural India save more in advance.

Timing. Although our sample is quite small, we can examine heterogeneity in our effects by the firstborn child's age to better understand how far in advance households change their saving and labor supply decisions. The increase in per capita savings starts becoming significant by the time the firstborn is age 10 (see Table 10). Note that the effect on savings is positive and meaningful even when the firstborn is under-5. We observe a similar pattern in Columns (4)-(6) of Table 8, which show that father's labor supply increases several years before the typical age at marriage.

Barriers to savings. A large literature in development economics shows that households in low-income countries face substantial barriers to savings (DellaVigna (2009), Karlan et al. (2014), Kremer et al. (2019)). These barriers arise due to income constraints, information and knowledge gaps, poor access to formal savings products, and behavioral constraints, such as present-bias, lack of self-control, and limited memory and attention, that may in turn be exacerbated by poverty. We believe that there are several reasons why an average rural Indian household may be better able to overcome these constraints in the context of dowry, as implied by our results, relative to investing in preventive health care, for instance.

First, households may have a better understanding of the costs and returns of saving for dowry relative to other saving motives. The social and economic costs that ensue from the inability of parents to get a daughter married to an appropriate groom by a certain age due to the lack of sufficient dowry tend to be quite high (Bloch and Rao (2002), Sekhri and Storeygard (2014), Maertens (2013)). Moreover, parents have relatively good information on the prevailing rate of dowry in the relevant marriage market; and there is typically no uncertainty about whether or not they will need to pay dowry. Second, mental accounting for dowry-related saving may be easier than for other financial decisions. Third, intra-household disagreement on saving for dowry is likely to be low as all household members may bear the cost in case of insufficient dowry payments for a female household member. For instance, a younger sister's marriage prospects may worsen if her older sister remains unmarried or obtains a worse marriage market match due to insufficient dowry. A brother may have to make future transfers to his sister's in-laws to prevent her from being abused, or bear the responsibility of a sister's living expenses if her marriage breaks down, due to insufficient dowry. Fourth, given that marriage is a frequent and elaborate public event in rural India, each wedding in a household's social network is likely to serve as a "reminder" to save for dowry, thereby overcoming the limited attention bias that could lead to under-saving in other circumstances. Fifth, peer influence and social image concerns are also likely to be stronger around long-standing cultural institutions, such as, dowry.

Although an average household in our sample is able to overcome barriers to savings in case of dowry, we find that these constraints are binding for the poorest households in our sample. As Tables 8 and 10 show, our savings and labor supply results are driven by households that are above the poverty line. The extremely poor households, i.e., those below the poverty line, are unable to significantly increase their savings for dowry in advance. This implies that the income or behavioral constraints do matter for these households.

[42] The average likelihood of borrowing in the year before the survey is 19 percent for any loan and 1.49 percent for a loan for marriage or ornaments.

Table 9
Impact of expected dowry on borrowing.

A. Borrowed last year = 1	All	All	Above poverty line	Below poverty line
	(1)	(2)	(3)	(4)
*Firstborn girl * Expected dowry*	0.02**	0.03	0.02	0.03
	[0.01]	[0.02]	[0.04]	[0.08]
Expected dowry	0.04			
	[0.03]			
Firstborn girl	−0.08	−0.23**	−0.23	−0.21
	[0.10]	[0.09]	[0.30]	[0.22]
N	2,846	2,846	1,463	1,176
Dep var mean for Firstborn boy	0.19	0.19	0.24	0.15
Caste*State*YOB FE		x	x	x
B. Amount borrowed in a year	All	Above poverty line	Below poverty line	
	(1)	(2)	(3)	
*Firstborn girl * Post* Expected dowry*	637.49	953.00*	45.79	
	[372.07]	[487.29]	[151.58]	
*Expected dowry * Post*	318.94**	259.50***	7.20	
	[149.02]	[82.81]	[122.20]	
*Firstborn girl * Post*	−1,642.62*	−2,530.33*	−278.76	
	(866.45)	(1,234.56)	(523.21)	
N	92,340	48,480	37,230	
Dep var mean for Firstborn boy	552.47	786.06	299.10	

NOTES: This table reports the coefficients corresponding to specifications (1) and (2) in Panel A and to specification (4) in Panel B. The sample is restricted to households where all children have the same mother. Each column corresponds to a different regression. The dependent variable is an indicator for borrowing any loan during 2007 in Panel A and the total amount borrowed in a year in Panel B. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Dep var mean is the mean of the dependent variable for firstborn boy households. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

Table 10
Impact on household per capita saving by poverty status and age of firstborn child.

Dependent variable:	Household per capita saving in 2007 (financial institutions + cash + interest earned)				
	Above poverty line (APL)	Below poverty line (BPL)	Firstborn ≤ 5	Firstborn ≤ 10	Firstborn ≤ 15
	(1)	(2)	(3)	(4)	(5)
*Firstborn girl * Expected dowry*	1881.66**	71.74	1353.38	1139.84**	707.71**
	[679.91]	[114.13]	[996.04]	[417.54]	[325.12]
Firstborn girl	−5419.96	−392.07	−7955.71	−4666.32	−1899.76
	[5895.24]	[406.53]	[5048.26]	[3293.17]	[3425.55]
N	1,461	1,172	795	1,510	2,328
Dep var mean for Firstborn boy	1415.33	135.28	836.10	856.67	669.87
CST Fixed effects	x	x	x	x	x

NOTES: Columns (1) and (2) report the coefficients for specification (2) separately for APL and BPL households. The BPL status is measured at the time of survey. Columns (3)–(5) split the sample by the age of the firstborn child. The sample is restricted to households where all children have the same mother. The dependent variable is the sum total of per capita savings in financial institutions, cash savings, and interest earned in 2007. Each column is a separate regression. *Firstborn girl* indicates that the first-born child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Dep var mean is the mean of the dependent variable for the firstborn boy households. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

Type of savings. The higher savings in FG families take the form of higher per capita formal saving in financial institutions. As Table 11 shows, we do not find a significant difference in the annual flow of jewelry saving (in precious stones and metals) among FG and FB families.[43] Similarly, there are no significant differences in the flow of saving in livestock (although the coefficient is positive) and saving in durable goods. One may be concerned that the lack of significant effects on these flow measures is due to the fact that purchases of durable goods are relatively infrequent. Therefore, we also examine if using the *stock* versions of the saving variables in Table 11 yields different results. The interaction coefficients remain insignificant, however (Table A.13). This pattern of saving behavior is consistent with greater access to the banking system in rural India and the less liquid nature of jewelry relative to cash savings in bank accounts during our study period.

Magnitudes. Back-of-the-envelope calculations suggest that FG families accumulate about INR 82,817 per daughter by the time she is 15 years old, the median age at marriage in our sample.[44] According

[43] This is true irrespective of whether the family has only one child or more children.

[44] The average household size in our sample is 5.78. Thus, for the mean level of expected dowry (=INR 26,120), FG families accumulate INR 89,653 more than FB families over 15 years (= 395.89 * 2.612 * 5.78 * 15). Since FG families have 1.09 more girls than FB families, this implies that FG families save roughly INR 82,251 per daughter.

Table 11
Impact of expected dowry on the flow of other types of savings.

Dependent variable:	Livestock	Market investments	Jewelry	Durable goods
	(1)	(2)	(3)	(4)
*Firstborn girl * Expected dowry*	331.08	−93.18	−5.51	−67.99
	[213.72]	[60.52]	[181.92]	[46.58]
Firstborn girl	1533.96	391.50	147.82	401.11
	[969.48]	[559.45]	[950.41]	[276.28]
N	1,623	2,845	2,079	2,844
Dep var mean for Firstborn boy	−72.32	150.11	158.26	334.76

NOTES: This table reports the coefficients corresponding to specification (2) for different types of saving. The sample is restricted to households where all children have the same mother. Each column is a separate regression. The dependent variables in each column are the flow of per capita household saving in livestock, market investments, jewelry, and durable goods. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Dep var mean is the mean of the dependent variable for firstborn boy households. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

to 2005 IHDS data, an average bridal family spends INR 75,777 to INR 98,498 on a wedding in rural India. Thus, an average household appears to be able to save for the bulk of the wedding expenditure in advance. Any remaining shortfall is potentially met through borrowing during or close to the year of marriage.

The average daily wage for agricultural casual labor in our data is INR 61 and, for non-agricultural casual labor, it is INR 91. Assuming that fathers are able to earn these wages for the extra days worked, their income increases by INR 610 to INR 910 each year relative to their income before the birth of their first child and relative to FB families. Although the increase in labor income is significant, it is smaller than the increase in savings that we observe. It is quite possible that parents have other sources of income, besides father's labor income, that we do not observe in our data, or that other members, e.g., brothers, may also increase their labor supply to supplement their fathers' earnings to finance the increased savings for their sister's dowry.

7. Robustness checks

In this section, we discuss several robustness checks that support our key findings.

7.1. Alternate definitions of expected dowry

In our main analysis, we proxy for dowry expectations with the average net dowry paid by brides or received by grooms from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Now, we examine the sensitivity of our estimates to alternative definitions of expected dowry.

First, we reconstruct the expected dowry variable by incorporating both religion and caste directly in the definition of the marriage market.[45] Specifically, we split Hindus by caste and use other religions as it is (i.e., our seven groups are: Hindu SCs, Hindu STs, Hindu OBCs, Hindu OCs, Muslims, Sikhs, Other religions) and then separately define expected dowry for these groups (while using state and year of birth as before).[46] Column (1) of Table A.14 shows that our savings results are robust to this alternate definition of expected dowry.

Second, instead of using the average of net dowries paid in marriages that occurred during the year of the child's birth (YOB) or the prior four years, in column (2) of Table A.14 we use the average of net dowries paid *around* the YOB of the child (i.e., during $YOB+2, YOB+1, YOB, YOB-1, YOB-2$). Once again, the interaction coefficient remains positive and statistically significant.

Third, we show the robustness of the main results using *median* dowry (instead of average dowry) to proxy for expected dowry in Table A.15. The point estimates of the interaction coefficient are similar in magnitude to Table 3 (our main table for the savings results), but we lose significance in column (6) of Table A.15. Table A.16 shows that results for father's labor supply are also robust to using the median to define expected dowry.

Finally, if we replace our expected dowry variable with expected dowry at the jati x State x Year level, we find qualitatively similar results, although we lose significance for the interaction terms. These results are available in Table A9 of the Online Appendix where we pool all jatis that have less than 100 observations into one jati. Although this is reassuring, using jati level dowry is not our preferred specification for the reasons outlined previously.

7.2. Missing observations

In 2006 REDS, we observe 17,401 marriages for which the year of marriage is available and is during 1986–2007. In the analysis so far, we have excluded marriages where data on both gifts given and received is missing (209 observations). Among the rest, while 8128 (47.28 percent) observations have information on both gifts, the remaining 9064 (52.72 percent) have one of them missing. In the latter case, when only one of the two is missing, we have calculated net dowry by assuming that the missing value equals zero. In doing so (i.e., by replacing missing data with zeros), we are primarily underestimating gifts from the groom's side, and in turn overestimating net dowry, since in 96.22 percent of the cases where one of the gifts is missing, the missing data is for gifts from the groom's side. Therefore, we test if our findings are driven by our treatment of missing data.[47] Reassuringly, our results remain the same if we construct expected dowry by only using marriages where both gifts are non-missing (see column (3) of Table A.14).

7.3. Expected gross marriage payments

Like most other papers in economics, we have modeled dowry as net dowry, following Becker (1981). However, gross payments from the bride's family may be more relevant in explaining savings as the bridal family must incur its share of the gross expenditure in advance and before receiving the payments from the groom's side during the marriage. We check how replacing net dowry with its two component variables, i.e., gross payments by the bride's and by the groom's family in specification (2) alters our estimates. Since payments by the groom

[45] In the main analysis, we incorporate religious heterogeneity by controlling for religion dummies in the regressions.

[46] Hinduism is the majority religion in India, and although other religions also exhibit caste, our sample size prevents us from splitting non-Hindus into further groupings by caste.

[47] We discuss missing observations in more detail in Appendix C.



Fig. A.1. Real marriage payments (in INR), by year of marriage. NOTES: The top figure plots the raw data on the real net dowry paid by the bride's family for all marriages in our data that took place during 1986–2007, by year of marriage. Each dot denotes a marriage. The bottom graph distinguishes between the real value of the gifts from the bride's family and from the groom's family.

are much smaller than those by the bride, we do not expect this to matter. Table A.17 confirms our intuition. The coefficient of *Firstborn girl * Expected gross payment by bride* continues to be positive and significant, and similar in magnitude to the coefficient of *Firstborn girl * Expected net dowry* in Table 3.

7.4. Dropping one state at a time

We also test the sensitivity of our savings and father's labor supply results to the inclusion of any one particular state by re-estimating specifications (2) and (3) while dropping one state at a time. The coefficients for the main interaction terms are reported in Table A.18. It is reassuring that the magnitude of the coefficients remains similar to our previous estimates. Moreover, out of 34 regressions, we lose significance only in three cases.

7.5. Placebo tests

Lastly, to further ensure that our expected dowry variable is not capturing the effect of other variables, we construct alternative average measures at the Caste x State x Year level for six other variables: total household expenditure, per capita household expenditure, father's years of education, mother's years of education, acres of land owned by the household, and per capita acres of land owned by the household, and then run placebo regressions by replacing the expected dowry variable in our specification (1) and (2) with these measures. In all cases, the coefficient of the interaction term in Table A.19 is insignificant. This lends support to our argument that our results are indeed capturing

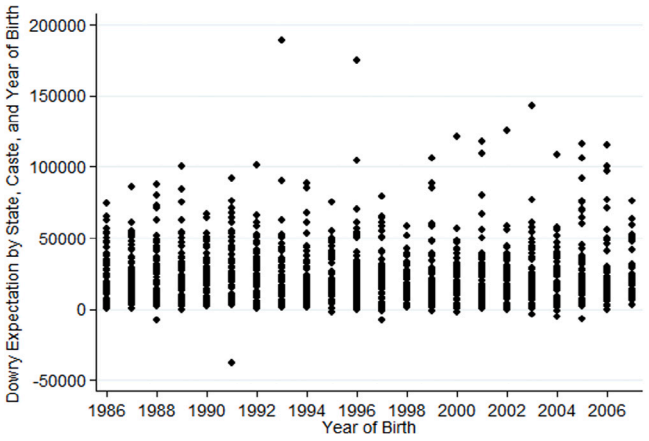


Fig. A.2. Our expected dowry variable (in INR), by year of birth of the firstborn child. NOTES: This figure plots our expected dowry variable, by year of birth of the firstborn child. Each dot denotes a caste–state cell. In this graph, we remove a few outlier cells where expected dowry is greater than INR 200,000.

Table A.1
Composition of dowries in rural India, 2005 IHDS.

Item	% of households that respond "Usually given"
Utensils	80.42
Bedding/mattress	70.94
Gold	70.78
Silver	63.85
Watch	62.41
Furniture	48.89
Cash	41.03
Pressure cooker	35.12
Sewing machine	21.71
TV	20.42
Bicycle	16.60
Mixer or grinder	15.70
Fridge	9.81
Livestock	8.78
Scooter or motorcycle	7.49
Land	1.04
Car	0.44
Tractor	0.32

NOTES: This table shows the percentage of rural households that respond "Usually" to the following question: *Generally in your community for a family like yours, what are the kind of things that are given as gifts at the time of the daughter's marriage?* in 2005 IHDS.

Table A.2
Does our expected dowry variable predict actual dowry?.

Dependent variable:	Actual dowry paid			
	(1)	(2)	(3)	(4)
Expected dowry	0.537*** [0.099]	0.530*** [0.090]	0.187** [0.095]	0.144** [0.056]
N	19,664	19,664	19,664	19,664
X_i		x	x	x
Caste FE			x	x
State FE			x	x
YOB FE			x	x
State FE*YOB FE				x
Caste FE*YOB FE				x
Caste FE*State FE				x

NOTES: In this table, we regress the net dowry paid or received by an individual on our expected dowry variable. $\mathbf{X}_i$ controls for indicators of household religion. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

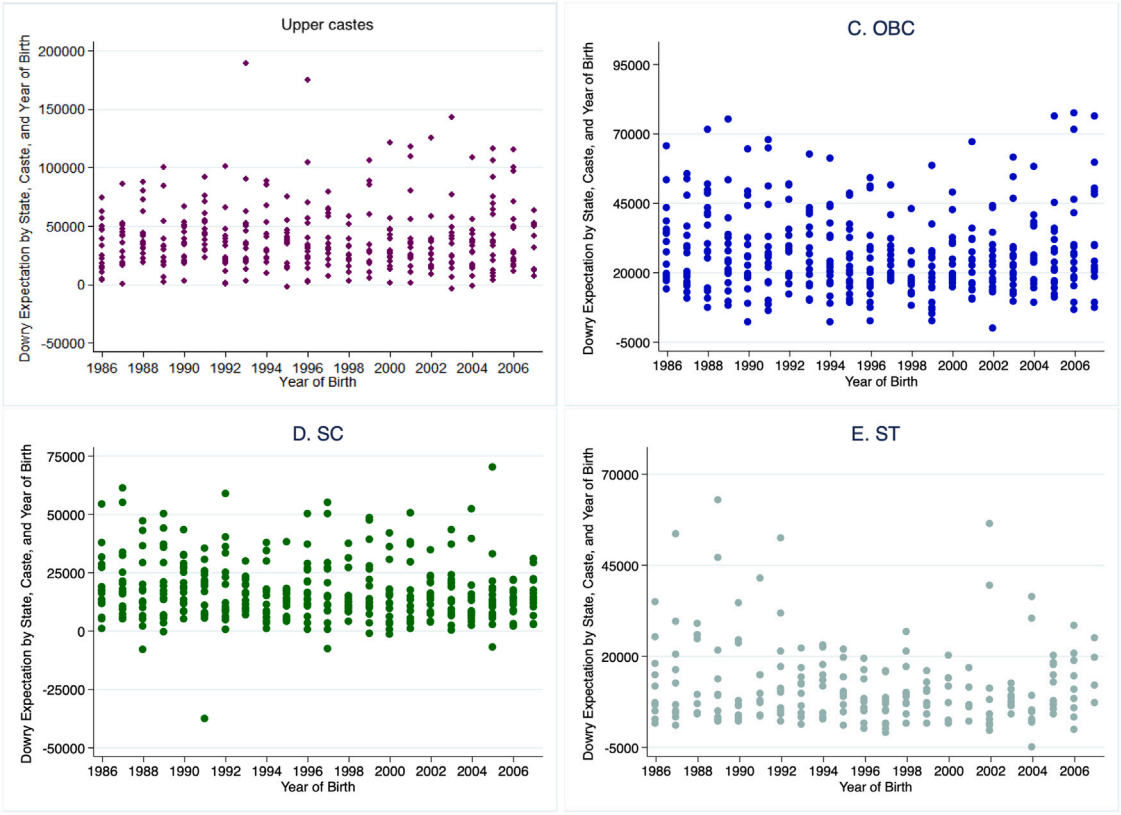


Fig. A.3. Our expected dowry variable (in INR), by year of birth of the firstborn child and caste. NOTES: This figure plots our expected dowry variable by year of birth of the firstborn child, separately for each caste group. OBC, SC, and ST stand for other backward classes, scheduled castes, and scheduled tribes, respectively. In this graph, we remove a few outlier cells where expected dowry is greater than INR 200,000.

the impact of expected dowry and not of other variables that may be correlated with $Dowry_{cst}$.

7.6. Inference

To take into account the fact that our expected dowry variable is a generated regressor, we also estimate standard errors that bootstrap both the construction of our dowry variable as well as the regressions together and by drawing separate bootstrap samples for the two steps. In both cases, our coefficients remain largely significant. These results are available in Table A10 of the Online Appendix.

8. Conclusion

A large literature has shown that culture matters for economic outcomes. More narrowly, there is a small but emerging body of work that highlights the importance of marriage-related cultural norms and institutions for households' decision-making. Despite the wide prevalence of bride-to-groom marriage payments, i.e., dowries, in several developing countries, economists have not directly investigated their impacts on financial and childbearing decisions, and on human capital investments in children. This is due to both the lack of data on dowry and the lack of a credible identification strategy. In this paper, we make use of an under-utilized source of nationally representative dowry data from India and propose a novel estimation strategy to examine the impact of future dowry payments on current outcomes.

We find several important results. This is the first paper, that we are aware of, that proposes an alternative motive for savings behavior in dowry-paying societies: the prospect of lump-sum dowry expense induces firstborn-girl families to start saving more in advance relative to firstborn-boy families in rural India. The ability of parents to overcome

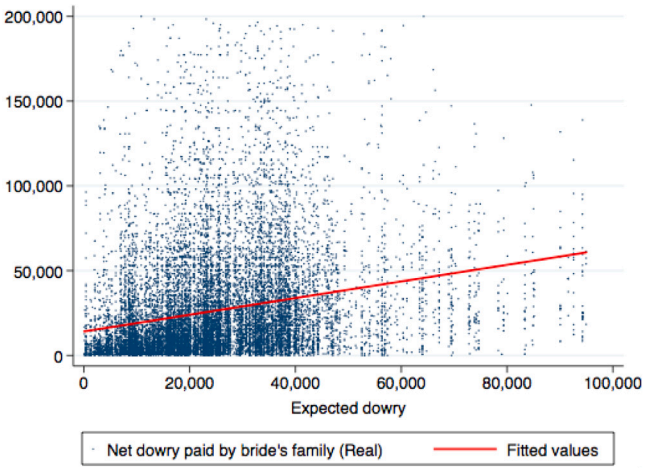


Fig. A.4. Correlation between expected and actual dowry. NOTES: This figure shows how our expected dowry variable (defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years) covaries with actual dowry. This figure is based on the sample of individuals in 2006 REDS whose year of birth and year of marriage span our dowry data. The sample is restricted to observations where expected dowry $\in$ (0, 100000) and actual dowry $\in$ (0, 200000) to prevent outliers from biasing the figure.

barriers to savings, especially behavioral biases such as imperfect self-control, in the context of saving for dowry, suggests that it is crucial to take into account the cultural context while designing policies that seek to affect saving behavior. We find that the increased savings take the form of formal savings in financial institutions and not savings in jewelry or precious metals that are traditionally considered an integral

Table A.3
Descriptive statistics on wedding expenditure from the 2005 India Human Development Survey.

	Panel A: By caste									
	Urban					Rural				
	Brahmin (1)	OBC (2)	SC (3)	ST (4)	Others (5)	Brahmin (6)	OBC (7)	SC (8)	ST (9)	Others (10)
Expenditure by bride's family	186,683	120,818	96,038	74,689	165,710	140,575	89,973	64,194	35,007	120,038
Expenditure by groom's family	128,166	74,673	63,616	58,691	108,188	92,365	56,380	43,539	29,176	74,162
Difference	58,617	46,145	32,423	15,997	57,522	48,210	33,589	20,655	5,835	45,875
N	1,313	5,452	2,322	499	4,956	1,108	10,834	6,011	2,939	6,118

	Panel B: By religion							
	Urban				Rural			
	Hindu (1)	Muslim (2)	Christian (3)	Sikh (4)	Hindu (5)	Muslim (6)	Christian (7)	Sikh (8)
Expenditure by bride's family	137,992	119,959	143,553	194,341	84,014	100,849	103,211	161,513
Expenditure by groom's family	90,649	76,515	66,274	132,988	55,309	57,894	48,767	107,471
Difference	47,350	43,444	77,278	61,353	28,706	42,956	54,443	54,042
N	11,286	2,215	514	258	22,239	2,573	862	732

NOTES: This table provides means of wedding expenditure in the 2005 India Human Development Survey (IHDS). The survey asks "At the time of the marriage in your community (*jati*) for a family like yours, how much money is usually spent by the girl(boy)'s family?" The IHDS dataset has five broad social groups: (1) Brahmin (2) OBC (3) SC (4) ST (5) Others.

Table A.4
Summary Statistics for Outcome Variables.

Variable	N	Mean	Standard deviation	Min	Max
	(1)	(2)	(3)	(4)	(5)
Saving in financial institutions	3,072	573.11	4,735.66	−100,000	113,447.50
Saving in financial institutions + Cash	3,072	477.59	5,178.75	−100,800	112,947.50
Saving in financial institutions + Cash + Interest earned	3,072	647.50	5,556.19	−100,395.04	113,007.41
Father's days worked	71,283	160.14	125.46	0	365
Dummy for borrowed any loan during 2007	3,078	0.21	0.40	0	1
Total no. of children	3,078	2.42	1.25	1	9
Total no. of girls	3,078	1.08	1.01	0	6
Total no. of boys	3,078	1.34	0.91	0	6

NOTES: This table provides summary statistics for the outcome variables used in our regression analyses.

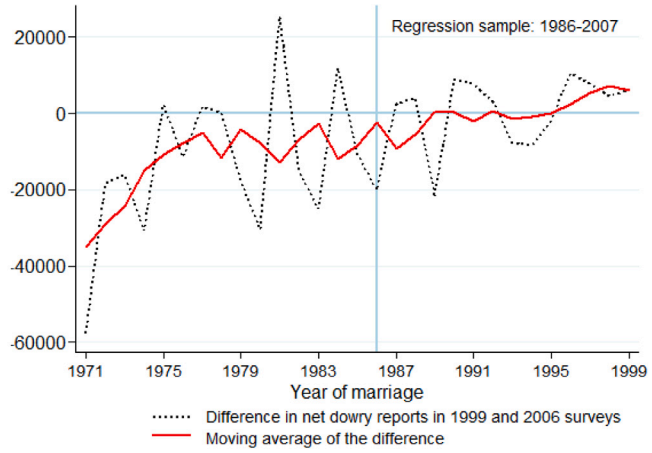


Fig. A.5. Recall bias in real dowry payments (in INR), by year of marriage. NOTES: The dashed line plots the difference between average real net dowry by year of marriage in the 1999 and 2006 rounds of REDS for marriages that took place during 1986–1999 (2006 minus 1999 amounts). The solid line plots the 5-year moving average of the difference. The vertical line denotes the earliest year included in our regression analysis.

part of the dowry in India. The higher savings that we find are financed through higher earnings; fathers of firstborn girls work more days in a year relative to FB fathers as expected dowry burden goes up. We also find that brothers in FG families decrease their years of schooling as expected dowry expenditure goes up, suggesting that they may also be working more in order to finance higher savings for their sisters' dowry (these results are available upon request).

In the context of India, it is frequently claimed that dowry is an underlying cause of son preference, male-biased fertility, and discrimination against girls. Contrary to this, we find that dowry is unlikely to be a root cause of son-preferring behaviors in India.

Future research should examine if the results of this paper also hold true for urban India, for a more recent time period, and for other dowry-paying societies. How alternate marriage market institutions, such as bride price, affect households' financial decisions also remains a fruitful area for future research. ■

CRediT authorship contribution statement

S. Anukriti: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing. **Sungoh Kwon:** Methodology, Software, Formal analysis, Visualization. **Nishith Prakash:** Conceptualization, Methodology, Formal analysis, Writing.

Data availability

The authors do not have permission to share data.

Appendix A. Appendix figures and tables

see Figs. A.1–A.7, Tables A.1 and A.19.

Appendix B. Sample-selection criteria

In this appendix, we describe our REDS sample selection criteria and the reasons why we need to impose them. First note that we do not have panel data on household savings. We only know how much a household saved during the year before the survey. To test if

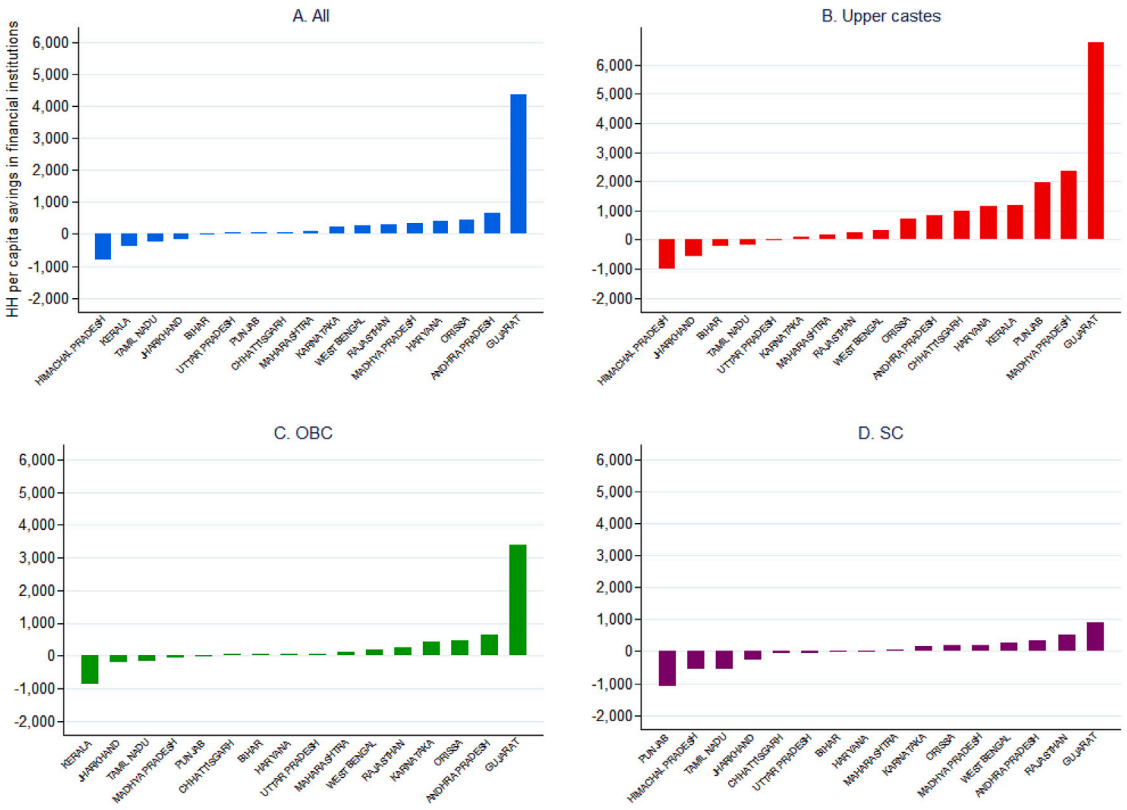


Fig. A.6. State-wise average per capita savings in financial institutions (in INR). NOTES: This figure shows how average per capita flow of household savings in financial institutions, cash savings, and interest earned during 2007 (in 2005 INR) varies across states within each caste group. OBC and SC stand for other backward classes and scheduled castes, respectively.

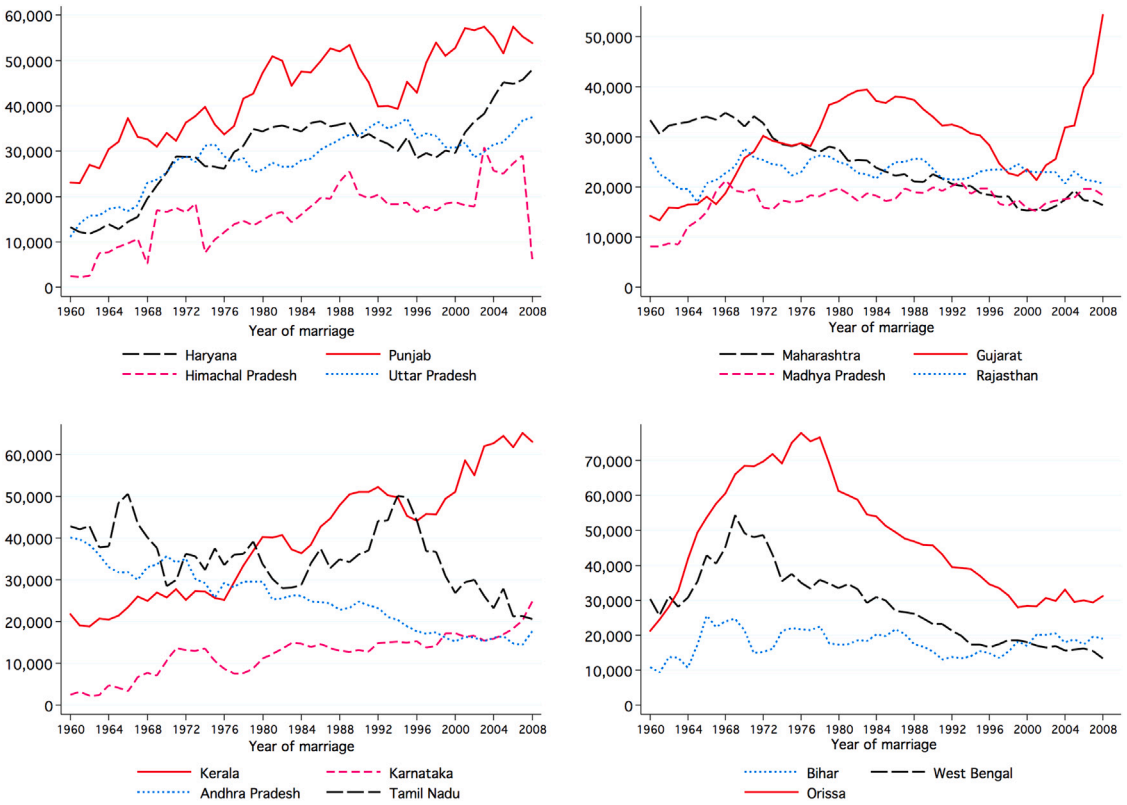


Fig. A.7. Trends in real dowry payments (in Rupees), by state and year of marriage. NOTES: This figure plots the 5-year moving unweighted average of real net dowry paid by the bride by year of marriage across states.

Table A.5
Dowry expense and the number of daughters.

Dependent variable:	Total dowry paid		Dowry paid per married daughter		Total dowry paid -Total dowry received	
	(1)	(2)	(3)	(4)	(5)	(6)
No. of married daughters	29,467.02*** [4,858.57]	30,929.49*** [4,567.83]	−1,637.66 [1,331.35]	−758.47 [975.88]		
Net no. of married daughters					12,825.20*** [2,288.45]	28,184.69** [11,212.55]
N	3,455	3,455	3,455	3,455	4,058	4,057
Caste FE		x		x		x
Religion FE		x		x		x
State FE		x		x		x

NOTES: In columns (1) and (2), we regress the total net dowry paid by the parents on the number of married daughters. In columns (3) and (4), we regress the net dowry paid per married daughter (defined as the total dowry paid by parents divided by the number of daughters) by the parents on the number of married daughters. In columns (5) and (6), we regress the total net dowry paid by the parents minus the total net dowry received by the parents on the net number of married daughters (= number of married daughters - number of married sons). Columns (2), (4), and (6) also control for fixed effects for caste, religion, and state. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

Table A.6
Effects on household per capita saving in financial institutions: Gradually adding fixed effects.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
*Firstborn girl * Expected dowry*	−18.499	4.888	31.328	58.159	50.885	61.393	350.138*	419.244**	556.246*
	(114.494)	(99.852)	(111.277)	(126.133)	(121.251)	(162.986)	(190.602)	(190.016)	(271.315)
Expected dowry	132.781	−61.710	65.782	95.694	235.628	221.049	78.864	60.857	
	(107.702)	(255.374)	(160.598)	(341.418)	(314.434)	(321.656)	(319.867)	(306.283)	
Firstborn girl	−309.800	−169.534	−202.498	−269.8	−292.774	−597.086	−1,605.135	−406.126	−639.849
	(296.990)	(184.202)	(208.785)	(228.960)	(239.105)	(430.449)	(1,025.921)	(1,728.706)	(3,342.154)
N	2,840	2,840	2,840	2,840	2,840	2,840	2,840	2,840	2,840
X_i	x	x	x	x	x	x	x	x	x
YOB FE		x	x	x	x	x	x	x	x
Caste FE		x	x	x	x	x	x	x	x
State FE		x	x	x	x	x	x	x	x
Caste FE*State FE			x	x	x	x	x	x	x
State FE*YOB FE				x	x	x	x	x	x
Caste FE*YOB FE					x	x	x	x	x
Firstgirl*Caste FE						x	x	x	x
Firstgirl*State FE							x	x	x
Firstgirl*YOB FE								x	x
Caste*State*YOB FE									x

NOTES: This table reports the coefficients corresponding to specifications (1) and (2) estimated for households where all children have the same mother. Each column is a separate regression. The dependent variable is total household saving in financial institutions. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Caste refers to indicators for SC, ST, OBC, and upper castes. YOB refers to the year of birth of the firstborn child. State refers to the state of residence at the time of survey. $\mathbf{X}_i$ controls for parents' age, schooling, religion, and month of survey. Standard errors in parentheses are clustered by state. *** 1%, ** 5%, * 10%.

Table A.7
Flow of Household per capita saving, by religion.

Dependent variable:	Household per capita saving in 2007					
	Hindus			Non-Hindus		
	Saving in financial institutions (1)	+ cash saving (2)	+ interest earned (3)	Saving in financial institutions (4)	+ cash saving (5)	+ interest earned (6)
*Firstborn girl * Expected dowry*	799.93** [366.70]	683.25* [343.17]	889.28* [447.13]	145.25 [841.86]	236.60 [837.35]	−17.20 [961.95]
Firstborn girl	−2,492.21 [2,992.07]	−2,847.52 [2,761.29]	−4,363.68 [3,278.80]	2,874.54 [5,678.14]	919.26 [5,527.69]	398.76 [6,368.59]
N	2,521			319		

NOTES: This table reports the coefficients corresponding to specification (2) estimated for households where all children have the same mother, separately for Hindus and non-Hindus. Each column is a separate regression. The dependent variable is total per capita household saving in financial institutions in columns (1) and (4), plus per capita cash saving in columns (2) and (5), and plus per capita interest earning in columns (3) and (6). *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

households are saving to finance future dowry for their children, we therefore need to focus on households that have unmarried individuals at the time of survey. Out of 8659 households in our REDS data, we drop 2089 households that either have no unmarried member or the unmarried members are all older than 18.[48] We drop households where

[48] The legal minimum age at marriage for women in India is 18.

Table A.8
Placebo test using expected dowry in the same region but *NOT* the same caste as the child.

Dependent variable:	Household per capita saving in 2007					
	Saving in financial institutions		Plus cash saving		Plus interest earned	
	(1)	(2)	(3)	(4)	(5)	(6)
*Firstborn girl * Placebo dowry*	−575.550	−764.545	−453.438	−559.185	−502.528	−620.974
	(669.016)	(868.222)	(731.196)	(978.721)	(733.988)	(968.643)
Placebo dowry	−562.316		−400.331		−483.330	
	(786.008)		(756.650)		(714.586)	
Firstborn girl	6,212.808	8,081.140	4,534.325	5,665.818	4,622.314	5,804.051
	(6,060.427)	(8,111.954)	(6,539.125)	(8,766.876)	(6,528.960)	(8,783.470)
N	2,840	2,840	2,840	2,840	2,840	2,840
Interview Month	x	x	x	x	x	x
Parents' Education/Age	x	x	x	x	x	x
Caste FE	x	x	x	x	x	x
YOB FE	x	x	x	x	x	x
State FE	x	x	x	x	x	x
State*YOB FE	x	x	x	x	x	x
Caste*YOB FE	x	x	x	x	x	x
Caste*State FE	x	x	x	x	x	x
Firstborn girl*YOB FE	x	x	x	x	x	x
Firstborn girl*State FE	x	x	x	x	x	x
Firstborn girl*Caste FE	x	x	x	x	x	x
Caste*State*YOB FE		x		x		x

NOTES: This table reports the coefficients from specifications (1) and (2) estimated for households where all children have the same mother. Each column is a separate regression. The dependent variable is the flow of per capita household saving in financial institutions in columns (1) and (2), plus per capita cash saving in columns (3) and (4), and plus per capita interest earning in the last two columns. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Caste refers to indicators for SC, ST, OBC, and upper castes. YOB refers to the year of birth of the firstborn child. State refers to the state of residence at the time of survey. $\mathbf{X}_i$ controls for parents' age, schooling, religion, and month of survey. Standard errors in brackets are clustered by state and wild-cluster bootstrapped errors by state are in parentheses. *** 1%, ** 5%, * 10%.

Table A.9
One-child families.

Dependent Variable:	Household per capita saving in 2007					
	Saving in financial institutions		Plus cash saving		Plus interest earned	
	(1)	(2)	(3)	(4)	(5)	(6)
*Firstborn girl * Expected dowry*	3366.41*	3506.17*	3528.10*	3528.16*	3714.13*	3717.74*
	[1821.46]	[1882.99]	[1787.73]	[1996.05]	[1752.35]	[1939.27]
Expected dowry	−1608.42*		−2930.04*		−2892.84*	
	[898.37]		[1473.94]		[1457.44]	
Firstborn girl	−16787.47*	−18631.00**	−19223.79**	−19646.58**	−21038.68**	−21556.61**
	[7941.73]	[8321.80]	[7644.09]	[8976.05]	[7766.93]	[8754.37]
N	671					
Dep var mean for Firstborn boy	840.38		628.36		801.81	
Caste*State*YOB FE		x		x		x

NOTES: This table reports the coefficients corresponding to specifications (1) and (2) estimated for households that have only one child. Each column is a separate regression. The dependent variable is total per capita household saving in financial institutions in columns (1) and (2), plus per capita cash saving in columns (3) and (4), and plus per capita interest earning in the last two columns. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Caste refers to indicators for SC, ST, OBC, and upper castes. YOB refers to the year of birth of the firstborn child. State refers to the state of residence at the time of survey. $\mathbf{X}_i$ controls for parents' age, schooling, religion, and month of survey. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

the unmarried individuals are older than 18 as these individuals are likely to be off the marriage market in our context, i.e., rural India. This leaves us with 6570 households that have at least one unmarried member who is 18 years of age or younger.

Next, to implement our identification strategy, we need to identify firstborn children. REDS does not report the birth order of household members, so we impute it by first linking individuals with their mothers and then assigning birth order by ranking siblings by age. We can only identify an individual's mother (and hence his/her siblings) if the mother resides in the same household. Therefore, we drop 406 households where the mother of a household member in our sample is not co-resident (e.g., because she might have died). This leaves us with 6164 households.

Among this sample, there are 727 households where multiple mothers and their children are present. Essentially, these are non-nuclear families (e.g., two brothers living together with their spouses and children) and, as such, they comprise multiple firstborn children from different mothers. We drop these from our sample because our outcome

Table A.10
Impact of expected dowry on fertility and sex ratio, REDS data.

	(1)	(2)	(3)	(4)	(5)
Dependent variable:	Number of births				
*Firstborn girl * Expected dowry*		0.085**	0.054*	0.055	0.044
		[0.031]	[0.029]	[0.042]	[0.045]
Firstborn girl	0.302***	0.080	0.123	0.119	0.132
	[0.060]	[0.126]	[0.124]	[0.161]	[0.156]
Expected dowry		−0.060	−0.022		
		[0.041]	[0.074]		
N	3,078	3,078	3,078	3,078	2,846
Dep var mean for Firstborn boy	2.285	2.285	2.285	2.285	2.285
Dependent variable:	Fraction sons (parity ≥ 2)				
*Firstborn girl * Expected dowry*		−0.004	−0.007	−0.008	−0.011
		[0.010]	[0.015]	[0.019]	[0.024]
Firstborn girl	0.073***	0.084*	0.101	0.102	0.113
	[0.021]	[0.043])	[0.059]	[0.073]	[0.079]
Expected dowry		0.009	−0.002		
		[0.007]	[0.029]		
N	2,342	2,342	2,342	2,342	2,174
Dep var mean for Firstborn boy	0.535	0.535	0.535	0.535	0.535
Caste FE + YOB FE + State FE			x	x	x
State*YOB FE			x	x	x
Caste*YOB FE			x	x	x
Caste*State FE			x	x	x
Caste*State*YOB FE				x	x
$\mathbf{X}_i$					x

NOTES: This table reports the coefficients corresponding to specification (3). Each column is a separate regression. The sample is restricted to households where all children have the same mother. The dependent variable is the proportion of male births among second and higher parity births. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Dep var mean is the mean of the dependent variable for firstborn-boy households. YOB refers to the year of birth of the firstborn child. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

Table A.11
Consumption.

	PC total expenditure		PC food expenditure		PC non-food expenditure	
	APL (1)	BPL (2)	APL (3)	BPL (4)	APL (5)	BPL (6)
*Firstborn girl * Expected dowry*	160.05	122.77	−6.13	53.10	158.30	83.85
	[708.13]	[333.11]	[81.45]	[181.74]	[671.57]	[223.88]
Firstborn girl	-7,402.23*	1,367.72	-3,017.97***	50.12	-2,465.83	1414.80
	[3,786.08]	[1,092.65]	[792.89]	[258.73]	[3,853.80]	[1029.46]
N	1,463	1,176	1,451	1,162	1,451	1,162

NOTES: This table reports the coefficients corresponding to specification (2) estimated for households where all children have the same mother, separately for above poverty line (APL) and below poverty line (BPL) families. Each column is a separate regression. The dependent variable is total per capita consumption expenditure in columns (1) and (2), per capita food expenditure in columns (3) and (4), and per capita non-food expenditure in columns (5) and (6). *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

variable (savings) is only available at the household level, whereas one of the key explanatory variables (sex of the firstborn child) varies at the sub-household level in non-nuclear families. The reasoning underlying our estimation strategy does not apply in case of non-nuclear families that have multiple earners and multiple parent–child combinations. Financial decision-making in joint families may be very different, and to that extent our findings about savings and labor supply do not apply to joint households.

This leaves us with 5437 households where all unmarried under-18 individuals have the same mother. Next we assign birth order by ranking siblings in terms of their age. However, this ranking will be biased if the firstborn is dead. So, we utilize mother interviews where they were asked if their first child is alive and, if alive, they were asked about the age of the oldest living child. Thus we cannot identify a firstborn child in our data if he/ she is not alive at the time of survey. We also drop households if the age of the oldest child reported by the mother does not match the age of any individual in the household roster[49] or if there are multiple firstborn children for the same mother (23 households). This leaves us with 3365 households.

[49] Note that this also means that we are excluding households where the firstborn child is not co-resident. Unfortunately, we have no way to identify such children in the data.

Table A.12
Impact on father's labor supply: Extensive and intensive margin.

Dependent Variable:	Father's days worked in a year					
	Extensive margin			Intensive margin		
	All	APL	BPL	All	APL	BPL
	(1)	(2)	(3)	(4)	(5)	(6)
*Firstborn girl * Post* Dowry*	0.009	0.015**	0.006	3.178***	3.333**	2.490
	(0.006)	(0.007)	(0.012)	(0.984)	(1.537)	(1.664)
*Dowry * Post*	0.004	0.002	−0.005	−1.785**	−2.651***	−1.021
	(0.006)	(0.008)	(0.013)	(0.717)	(0.884)	(1.276)
*Firstborn girl * Post*	−0.027	−0.049*	−0.028	−8.010**	−6.713	−8.221
	(0.022)	(0.027)	(0.035)	(3.112)	(6.175)	(5.687)
N	71,282	36,703	29,395	48,827	23,858	21,449

NOTES: This table reports the coefficients corresponding to specification (4). The sample is restricted to households where all children have the same mother. Each column corresponds to a different regression. In cols 1–3, the dependent variable is an indicator that equals one if father's number of days worked each year is positive and is zero otherwise. In cols 4–6, the dependent variable equals father's number of days worked each year if his annual number of days worked is positive. APL and BPL denote above poverty line and below poverty line, respectively.*Firstborn girl* indicates that the firstborn child of the household is female. Post indicates that the year of labor is later than the first child's year of birth. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Standard errors in parentheses are clustered by state. *** 1%, ** 5%, * 10%.

Table A.13
Stock of per capita household saving.

Dependent variable:	Per capita stock of saving in 2007			
	Livestock	Market investments	Jewelry	Durable goods
	(1)	(2)	(3)	(4)
A: Below poverty line				
*Firstborn girl * Expected dowry*	315.99	−558.80	−883.42	178.84
	[583.19]	[1,157.03]	[864.38]	[407.08]
N	546	833	1,176	1,175
B: Above poverty line				
*Firstborn girl * Expected dowry*	−717.60	−2,267.07	−1,088.76	−265.10
	[1,411.16]	[1,584.68]	[969.70]	[1,794.44]
N	962	1,102	1,463	1,462

NOTES: This table reports the coefficients corresponding to specification (2) for different types of saving. The sample is restricted to households where all children have the same mother. Each column is a separate regression. The dependent variables in each column are the stock of per capita household saving in livestock, market investments, jewelry, and durable goods. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Panel A uses the sample of below poverty line families, while panel B examines above poverty line households. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

Table A.14
Robustness.

Dependent variable:	Household per capita saving in 2007 (financial institutions + cash + interest earned)		
	Caste–religion	Around YOB	Non-missing obs.
	(1)	(2)	(3)
*Firstborn girl * Expected dowry*	623.55**	368.73*	637.40*
	[220.87]	[200.02]	[354.23]
Firstborn girl	−3228.32	−826.20	−3635.87**
	[2032.56]	[2359.00]	[1669.78]
N	2,836	2,837	2,610

NOTES: This table reports the coefficients corresponding to specification (2) estimated for households where all children have the same mother. Each column is a separate regression. The dependent variable is the sum total of per capita savings in financial institutions, cash savings, and interest earned in 2007. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the average dowry paid (received) by brides (grooms) from the same social group and state as the child and who married during the year of the child's birth or the prior four years. In column (1), we construct seven social groups based on the caste and religion. Specifically, we split Hindus by caste and use other religions as it is (i.e., Hindu SCs, Hindu STs, Hindu OBCs, Hindu OCs, Muslims, Sikhs, Other religions). In column (2), expected dowry is defined using marriages around the year of birth (YOB) of the child (i.e., during YOB + 2, YOB + 1, YOB, YOB - 1, YOB - 2). In column (3), we construct expected dowry only using marriages where both gifts are non-missing. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

Table A.15
Savings: Using median dowry to proxy for expected dowry.

Dependent variable:	Per Capita Household Saving in 2007					
	Saving in financial institutions		Plus cash saving		Plus interest earned	
	(1)	(2)	(3)	(4)	(5)	(6)
*Firstborn girl * Expected dowry*	507.76**	524.60**	500.14**	488.13*	627.44**	648.63
	[174.49]	[240.32]	[188.95]	[264.50]	[266.96]	[393.25]
Expected dowry	42.75		−229.43		−218.03	
	[217.73]		[258.55]		[271.38]	
Firstborn girl	−152.45	270.08	−869.37	−460.25	−1,666.66	−1,420.38
	[1,500.31]	[2,914.30]	[1,520.13]	[2,909.32]	[1,836.77]	[3,376.16]
N	2,840					
$\mathbf{X}_i$	x	x	x	x	x	x
Caste FE	x	x	x	x	x	x
YOB FE	x	x	x	x	x	x
State FE	x	x	x	x	x	x
State*YOB FE	x	x	x	x	x	x
Caste*YOB FE	x	x	x	x	x	x
Caste*State FE	x	x	x	x	x	x
Firstborn girl*YOB FE	x	x	x	x	x	x
Firstborn girl*State FE	x	x	x	x	x	x
Firstborn girl*Caste FE	x	x	x	x	x	x
Caste*State*YOB FE		x		x		x

NOTES: This table reports the coefficients corresponding to specifications (1) and (2) estimated for households where all children have the same mother. Each column is a separate regression. The dependent variable is per capita household saving in financial institutions in columns (1) and (2), plus per capita cash saving in columns (3) and (4), and plus per capita interest earning in the last two columns. *Firstborn girl* indicates that the firstborn child of the household is female. Expected dowry (in INR 10,000) for a female (male) child is defined as the *median* net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Caste refers to indicators for SC, ST, OBC, and upper castes. YOB refers to the year of birth of the firstborn child. State refers to the state of residence at the time of survey. $\mathbf{X}_i$ controls for parents' age, schooling, religion, and month of survey. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

Table A.16
Father's labor supply: Using median dowry to proxy for expected dowry.

Dependent variable:	Father's days worked in a year		
	All	Above poverty line	Below poverty line
	(1)	(2)	(3)
*Firstborn girl * Post* Expected dowry*	5.83***	7.01***	4.75
	[1.75]	[1.89]	[4.58]
*Expected dowry * Post*	2.37	1.80	2.03
	[3.23]	[4.49]	[3.98]
*Firstborn girl * Post*	−11.12**	−13.09**	−13.90
	[4.62]	[5.55]	[9.76]
N	71,282	36,703	29,395

NOTES: This table reports the coefficients corresponding to specification (4). The sample is restricted to households where all children have the same mother. Each column corresponds to a different regression. The dependent variable is the number of days worked each year. *Firstborn girl* indicates that the firstborn child of the household is female. Post indicates that the year of labor is later than the first child's year of birth. Expected dowry (in INR 10,000) for a female (male) child is defined as the *median* net dowry paid (received) by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

Lastly, we drop 476 households where the interview was conducted in years other than 2008 to prevent any resulting bias due to omitted time trends or differential implementation of the survey. We are left with 2889 households.

Note that we are not the only ones to use some of these selection criteria. Several papers that rely upon household rosters often assign birth order by ranking children by age and impute relationships between household members by using information that is more readily available, for instance, using data on relationship with household head. Our other restrictions are either necessitated by our specific research question (e.g., we cannot analyze differences in marriage-related savings by child gender in households that have no unmarried members) or the underlying conceptual framework (e.g., the savings decision in joint families may have a different model than the one in nuclear households).

In any case, Table B.20 shows that these various sample-selection criteria do not significantly alter the composition of households in our final estimation sample in terms of caste, religion, BPL status, land ownership, household size, and household food and non-food expenditure. The final sample appears somewhat poorer than the starting sample,

Table A.17
Using *gross* (instead of net) marriage payments.

Dependent variable:	Household per capita saving in 2007 (financial institutions + cash + interest earned)	
	(1)	(2) FE for #children
*Firstborn girl * Expected gross payment by bride*	662.60*	656.62*
	[356.92]	[357.88]
*Firstborn girl * Expected gross payment by groom*	1066.17	1094.13
	[2129.19]	[2148.93]
Firstborn girl	−2572.92	−2539.77
	[3757.24]	[3787.71]
N	2,840	

NOTES: Instead of net dowry expectation in specification (2), here we use two gross dowry variables: *Expected gross (wedding) payment by bride* and *Expected gross (wedding) payment by groom*. Expected gross payment (in INR 10,000) by bride (groom) are defined as the average value of gifts given by brides (grooms) from the same caste and state as the child and who married during the year of the child's birth or the prior four years. Household data is restricted to households where all children have the same mother. Each column is a separate regression. Column (2) also controls for indicators for the number of children. *Firstborn girl* indicates that the firstborn child of the household is female. Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

Table A.18
Robustness check: Drop one state at a time.

Dependent variable:	PC Saving + Cash + Interest		Father's days worked	
	Coeff of *FG* Dowry*	N	Coeff of *FG* Post* Dowry*	N
Dropped state:	(1)	(2)	(3)	(4)
KERALA	617.54*	2,831	3.94**	71,057
	[304.40]		[1.61]	
KARNATAKA	615.26*	2,575	3.30*	64,608
	[322.66]		[1.83]	
MAHARASHTRA	644.16*	2,675	3.66**	67,053
	[316.86]		[1.69]	
GUJARAT	514.00	2,544	3.92**	63,807
	[322.70]		[1.64]	
MADHYA PRADESH	642.88*	2,548	4.44**	63,966
	[331.19]		[1.51]	
RAJASTHAN	618.36*	2,762	3.53*	69,342
	[306.96]		[1.70]	
HARYANA	439.11*	2,637	2.60	66,213
	[208.92]		[1.54]	
PUNJAB	807.30**	2,693	2.41	67,610
	[374.34]		[2.05]	
HIMACHAL PRADESH	547.89*	2,808	3.68**	70,488
	[309.65]		[1.68]	
UTTAR PRADESH	617.83	2,450	3.38*	61,562
	[355.07]		[1.91]	
BIHAR	675.91*	2,774	3.88**	69,638
	[328.51]		[1.66]	
WEST BENGAL	602.73*	2,695	3.89**	67,659
	[311.92]		[1.67]	
JHARKHAND	625.44*	2,771	3.88**	69,559
	[305.82]		[1.67]	
CHHATTISGARH	619.37*	2,702	3.35*	67,810
	[314.26]		[1.78]	
ORISSA	661.36*	2,711	3.95**	68,011
	[319.34]		[1.65]	
ANDHRA PRADESH	673.55*	2,569	3.37*	64,440
	[329.01]		[1.72]	
TAMIL NADU	636.45*	2,695	4.17**	67,689
	[316.14]		[1.57]	

NOTES: Standard errors in brackets are clustered by state. *** 1%, ** 5%, * 10%.

Table A.19
Placebo regressions.

Dependent variable:	Household Per Capita Saving in 2008		
	Saving in financialinstitutions (1)	Plus cashsaving (2)	Plus interestearned (3)
Panel A: Without CST FE			
1. Firstborn girl * Avg HH exp_{cst}	0.01 (0.04)	0.02 (0.04)	0.02 (0.04)
2. Firstborn girl * Avg PC HH exp_{cst}	0.12 (0.30)	0.19 (0.26)	0.22 (0.28)
3. Firstborn girl * Avg Father's edu_{cst}	25.84 (132.63)	61.62 (126.53)	68.60 (128.64)
4. Firstborn girl * Avg Mother's edu_{cst}	125.83 (127.23)	149.72 (146.78)	194.27 (154.22)
5. Firstborn girl * Avg Land $owned_{cst}$	−182.09 (172.06)	−123.50 (187.10)	−84.53 (199.28)
6. Firstborn girl * Avg PC Land $owned_{cst}$	−708.98 (1,421.66)	−107.29 (1,462.39)	213.90 (1,560.24)
N	2,840	2,840	2,840
Panel B: With CST FE			
1. Firstborn girl * Avg HH exp_{cst}	0.03 (0.06)	0.04 (0.05)	0.04 (0.05)
2. Firstborn girl * Avg PC HH exp_{cst}	0.14 (0.43)	0.23 (0.38)	0.27 (0.40)
3. Firstborn girl * Avg Father's edu_{cst}	−43.21 (218.49)	−15.35 (205.90)	−12.31 (208.81)
4. Firstborn girl * Avg Mother's edu_{cst}	95.09 (220.09)	112.22 (235.23)	157.64 (241.96)
5. Firstborn girl * Avg Land $owned_{cst}$	45.84 (206.91)	118.04 (187.99)	146.31 (197.46)
6. Firstborn girl * Avg PC Land $owned_{cst}$	445.35 (1,694.71)	1,176.72 (1,347.82)	1,407.86 (1,427.59)
N	2,840	2,840	2,840

NOTES: Each cell comes from a separate regression. Panels A and B correspond to specifications (1) and (2), respectively, except that the Expected dowry term is replaced with the state–caste–year average of another variable, such as HH expenditure. *** 1%, ** 5%, * 10%.

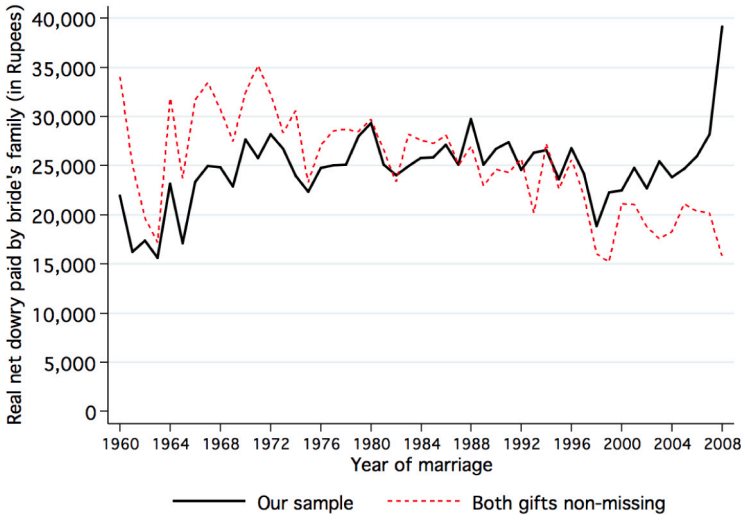


Fig. C.8. Trends in real marriage payments (in Rupees), by year of marriage. NOTES: This figure plots the raw unweighted average of the net dowry paid by the bride's family by year of marriage. The dashed line only uses observations that have non-missing information on gifts from both bride's and groom's sides. The solid line also includes observations where information on one of the gifts is missing and which we replace with a zero in calculating the net dowry.

but that should bias us towards finding no effects on savings if poorer households are more income-constrained.

Appendix C. Note on missing observations

In total, data on the year of marriage are available for 1960–2008 for 40,623 marriages. We exclude marriages where data on both gifts given and received are missing (1079) leaving us with 39,544 observations. While 18,275 (46 percent) observations have information on both gifts, the remaining 21,269 (54 percent) have one of them missing. In the latter case, when only one of the two is missing, we assume that the missing value equals zero. Note, however, that in 95 percent of the cases where one of the gifts is missing, the missing

Table B.20
Sample-selection criterion.

Variable	Starting sample		Keep unmarried <=18 members		Drop if momid=.		Drop if multiple mothers		Firstborn restrictions		Year of survey = 2008	
	N	Mean	N	Mean	N	Mean	N	Mean	N	Mean	N	Mean
SC	8,659	0.17	6,570	0.17	6,164	0.17	5,437	0.18	3,365	0.17	2,889	0.17
ST	8,659	0.08	6,570	0.08	6,164	0.08	5,437	0.08	3,365	0.08	2,889	0.09
OBC	8,659	0.47	6,570	0.47	6,164	0.47	5,437	0.47	3,365	0.47	2,889	0.46
Other castes	8,659	0.29	6,570	0.28	6,164	0.28	5,437	0.27	3,365	0.28	2,889	0.28
Hindu	8,659	0.89	6,570	0.89	6,164	0.89	5,437	0.89	3,365	0.88	2,889	0.89
Muslim	8,659	0.06	6,570	0.06	6,164	0.07	5,437	0.06	3,365	0.06	2,889	0.06
Sikh	8,659	0.03	6,570	0.03	6,164	0.03	5,437	0.03	3,365	0.04	2,889	0.04
Christian	8,659	0.01	6,570	0.01	6,164	0.01	5,437	0.01	3,365	0.02	2,889	0.01
BPL	8,084	0.43	6,151	0.42	5,774	0.42	5,085	0.44	3,149	0.44	2,687	0.47
Land owned	8,659	2.73	6,570	2.80	6,164	2.81	5,437	2.54	3,365	2.52	2,889	2.42
HH food exp	8,596	14723.55	6,520	15460.08	6,118	15514.69	5,397	14769.95	3,337	14871.66	2,861	14348.08
HH non-food exp	8,596	18243.03	6,520	19241.70	6,119	19359.51	5,398	17871.50	3,338	17640.28	2,862	16743.36
HH size	8,659	5.16	6,570	5.90	6,164	5.91	5,437	5.33	3,365	5.42	2,889	5.42

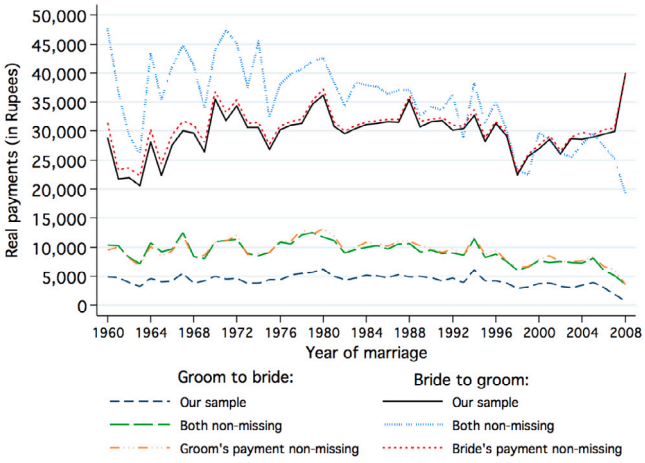


Fig. C.9. Trends in real marriage payments (in Rupees), by year of marriage. NOTES: This figure plots the raw unweighted average of the net dowry paid by the bride's family by year of marriage. The dashed line only uses observations that have non-missing information on gifts from both bride's and groom's sides. The solid line also includes observations where information on one of the gifts is missing and which we replace with a zero in calculating the net dowry.

data are for gifts from the groom's side. This implies that by replacing missing data with zeros we are primarily underestimating gifts from the groom's side, and in turn overestimating net dowry.

Fig. C.8 plots the trends in net dowry for our sample and for the sub-sample where both gifts are non-missing. The two lines are largely similar except in recent years for which we do not have a large enough sample size, suggesting that our analysis is not substantively affected by the treatment of missing data. This is not surprising since the bulk of the missing information is for groom's payments that are several orders of magnitude smaller than the bride's payments.

Fig. C.9 plots the trends for gross payments. In addition to the plots corresponding to Fig. C.8 (i.e., our sample and when both gifts are non-missing), a third set of lines plots average payments using non-missing data for each gift variable irrespective of whether the other gift variable is missing. As expected, for groom's payments, our sample means are lower (by about INR 5000) than those calculated using non-missing data. Average bride's payments are also somewhat smaller in our sample and the sample with non-missing bride's payments when compared to the sample where both gifts are non-missing.

Appendix D. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jdeveco.2021.102750.

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